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Ch. 4. Chemical equilibrium

REACTIONS rarely occur to completion. Once products form, they can reverse to regenerate the reactants, while reactants can also regenerate and produce products. Overall, chemical reactions proceed until reaching equilibrium. Chemical equilibrium and economic profitability are deeply linked in industrial chemistry. However, maximizing a chemical reaction's physical yield does not automatically maximize financial profit; instead, industrial engineers must balance equilibrium limits against operating costs, reaction rates, and material pricing. This chapter covers the idea and basic principles of chemical equilibrium. We will describe the role of the equilibrium constant, which gives insight into the reactant and product composition at equilibrium. We will also cover Le Châtelier's principle, which describes the direction in which a chemical reaction proceeds to recover the equilibrium state after equilibrium is altered.

4.1 Chemical Equilibrium

This section covers basic ideas about chemical equilibrium. First, we will introduce the concept of the forward and reverse reactions. Next, we will define the idea of equilibrium, based on the speed of the forward and reverse reactions.

Forward and reverse reactions In chemical reactions, reactants form products. We call this the *forward reaction*:

$$\text{reactants} \longrightarrow \text{products}$$

However, once the products form they can also generate reactants through the *reverse reaction*:

$$\text{reactants} \longleftarrow \text{products}$$

Equilibrium in terms of rate In chemical reactions, both the forward and the reverse reactions happen simultaneously, so when products form, reactants form as well. However, both the reverse ($r_{reverse}$) and forward ($r_{forward}$) reactions have different speeds. In the beginning, the forward reaction proceeds at a faster pace than the reverse so that the reaction advances. Once products form, the reverse reaction will start speeding up. A reaction reaches *chemical equilibrium* when both the forward and reverse reactions proceed at the same speed (see Figure 4.2).

$$r_{forward} = r_{reverse}$$

The term equilibrium in science refers to something remaining constant over time. Chemical reactions are normally written down with a double arrow that indicates equilibrium:

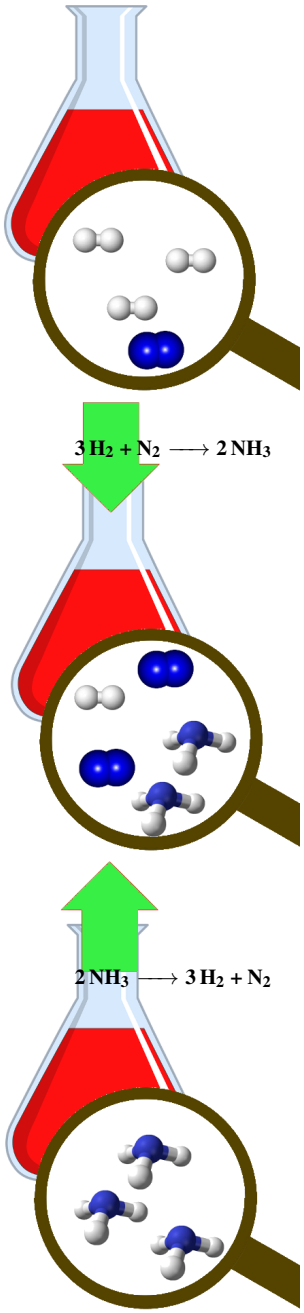
$$\text{reactants} \rightleftharpoons \text{products}$$

For example, the chemical process that forms ammonia from nitrogen and hydrogen can proceed using the forward and reverse reactions

$$3\,H_{2(g)} + N_{2(g)} \underset{\text{Reverse Reaction}}{\overset{\text{Forward Reaction}}{\rightleftharpoons}} 2\,NH_{3(g)}$$

Ammonia (NH_3) is a vital chemical because it feeds roughly 70% of the global population through agricultural fertilizers and serves as a core chemical building block for industry and clean energy.

Chemical equilibrium is a dynamic process. Even when the concentrations of reactants and products do not change with time, the reverse and forward reactions occur, and molecules still change at a microscopic level. A final thought: equilibrium and kinetics are different concepts. *Equilibrium* is governed by the thermodynamic characteristics of the reaction–the energy involved between reactants and products. In contrast, *kinetics* is related to the time it takes to reach equilibrium, and it is due to the rate laws that govern the forward and reverse processes, and ultimately the activation energy needed for reactants to overcome to become products.



▲The forward reaction goes from reactants to products whereas the reverse reaction goes from products to reactants.

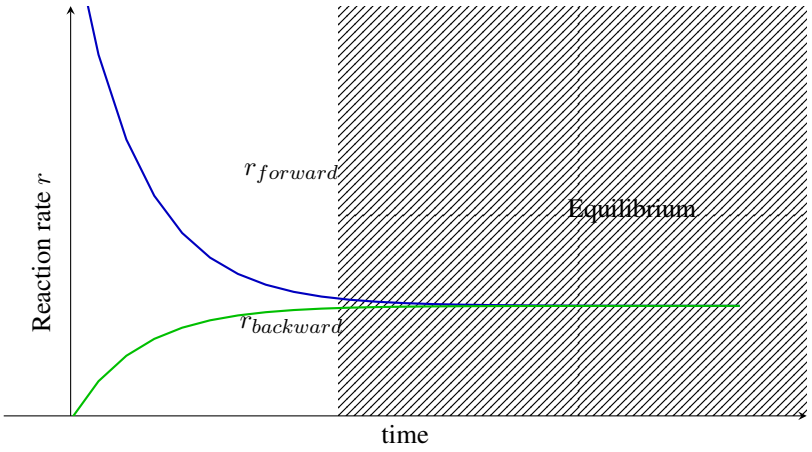


Figure 4.2 Changes with time of the rates for the forward and backward reaction, showing the point when equilibrium is reached.

Equilibrium in terms of concentration It is just a matter of time before a reaction reaches equilibrium. Initially, the forward process normally proceeds at high speed as products start to form. This is because initially there is an abundance of reactants. Once the products start to accumulate the forward process will start happening. This is because at that point there is an increasing amount of product molecules that can go back to reactants. Eventually, the forward and reverse rates become equal as the reaction reaches equilibrium. When the equilibrium has been reached, reactants and products have very specific concentrations that depend on temperature and are constant with time (see Figure 4.2). At the same time, a reaction in equilibrium contains a mixture of reactants and products.

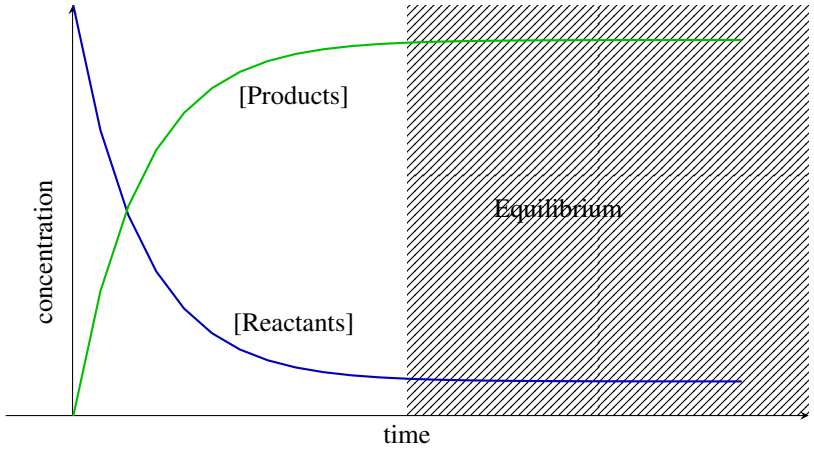


Figure 4.2 Change of the concentration of reactants and products with time showing the point when equilibrium is reached.

Sample Problem 1

Write down the forward and reverse reactions for:

$$CH_{4(g)} + 2\,O_{2(g)} \rightleftharpoons CO_{2(g)} + 2\,H_2O_{(g)}$$

SOLUTION

(a) The forward reaction is $CH_{4(g)} + 2\,O_{2(g)} \longrightarrow CO_{2(g)} + 2\,H_2O_{(g)}$ and the reverse $CH_{4(g)} + 2\,O_{2(g)} \longleftarrow CO_{2(g)} + 2\,H_2O_{(g)}$

STUDY CHECK

Write down the forward and reverse reactions for the reaction:

$$CO_{(g)} + O_{2(g)} \rightleftharpoons CO_{2(g)}$$

Answer: $CO_{(g)} + O_{2(g)} \longrightarrow CO_{2(g)}$ and $CO_{(g)} + O_{2(g)} \longleftarrow CO_{2(g)}$

4.2 The equilibrium constant

Reactions in equilibrium are characterized by an equilibrium constant. This next section will show how to interpret and calculate the value of the equilibrium constant of a reaction.

Equilibrium mixtures Imagine you start a chemical reaction. Initially, products will form while reactants disappear. At equilibrium, you will have a mixture of reactants and products with the forward and reverse processes happening at the same rate. Would an equilibrium mixture contain more reactants or more products? The equilibrium constant helps predict just that.

Table 4.1 K_c values for a set of different reactions at 298K

Reaction	K_c	Equilibrium mixture
$2\,NH_{3(g)} \rightleftharpoons N_{2(g)} + 3\,H_{2(g)}$	17	Products > Reactants
$H_{2(g)} + I_{2(g)} \rightleftharpoons 2\,HI_{(g)}$	50	Products > Reactants
$2\,SO_{3(g)} \rightleftharpoons 2\,SO_{2(g)} + O_{2(g)}$	0.3	Products < Reactants
$H_2O_{(l)} \rightleftharpoons H_2O_{(g)}$	0.2	Products < Reactants

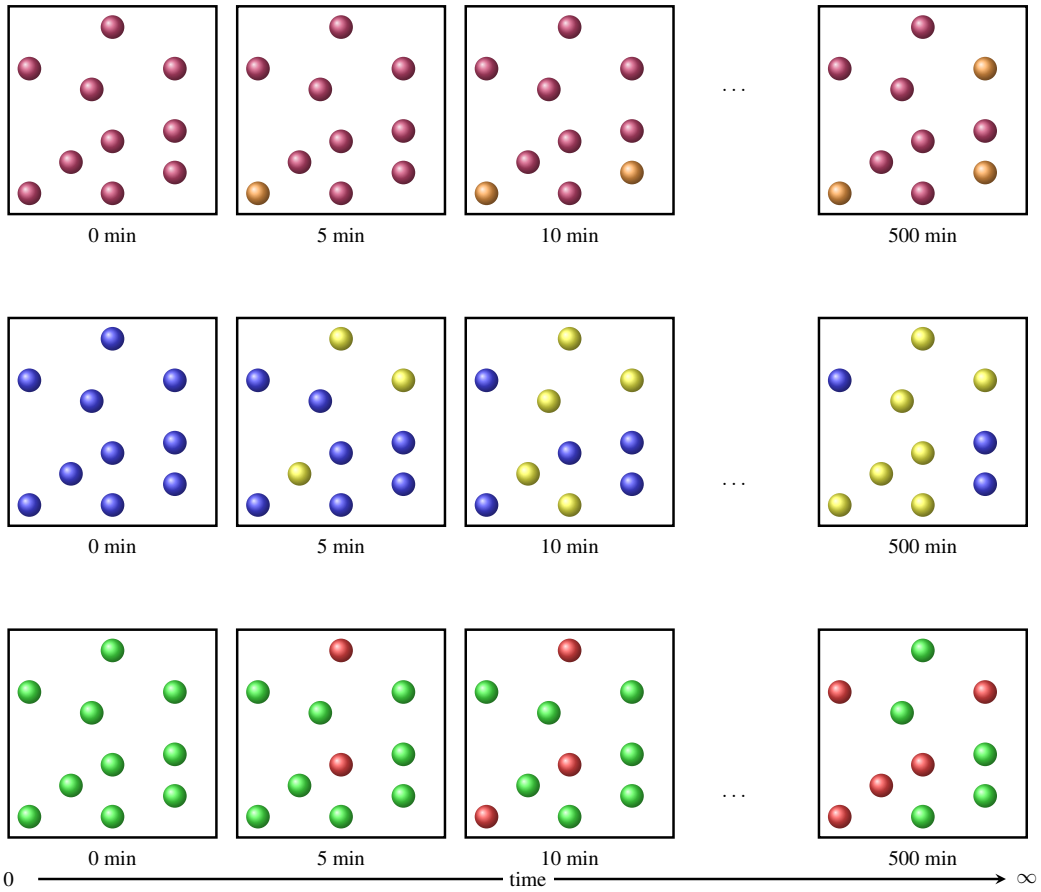


Figure 4.3 Three equilibrium situations. (Top Panel) At equilibrium, there are more reactants than products. (Central Panel) At equilibrium, there are more products than reactants. (Bottom Panel) At equilibrium, there are as many products as reactants.

The equilibrium constant of a reaction The equilibrium constant in terms of concentrations associated with a reaction K_c indicates whether reactants or products are more abundant at equilibrium. Each reaction has one K_c value that depends only on temperature and the subscript c represents concentration. On one hand, if K_c is larger than one there will be a larger concentration of products than reactants in the equilibrium mixture (see Figure 4.3). On the other hand, if K_c is smaller than one there will be a larger concentration of reactants than products in the equilibrium mixture. If K_c is close to one then both reactants and products will have the same concentration in the equilibrium mixture (see Table 4.1 for examples).



$K_c < 1$ $K_c \simeq 1$ $K_c > 1$

Sample Problem 2

The value of K_c for the following reaction at 300K is 4×10^{31}. Indicate whether the equilibrium mixture will contain mostly reactants, mostly products or both.

$$Cl_{2(g)} + H_{2(g)} \rightleftharpoons 2\,HCl_{(g)}$$

SOLUTION

As K_c is larger than one, an equilibrium mixture of Cl_2, H_2 and HCl will contain mainly products, that is will be mainly made of HCl.

STUDY CHECK

The value of K_c for $F_{2(g)} \rightleftharpoons 2\,F_{(g)}$ at 500K is 7×10^{-13}. Indicate whether the equilibrium mixture will contain mostly reactants, mostly products or both.

▶Answer: mostly reactants.

Equilibrium constant expression Let's consider a general equilibrium reaction in which A_1 and A_2 react to form A_3 and A_4

$$\underbrace{\nu_1 A_{1(Aq)} + \nu_2 A_{2(Aq)}}_{\text{Reactants}} \rightleftharpoons \underbrace{\nu_3 A_{3(Aq)} + \nu_4 A_{4(Aq)}}_{\text{Products}}$$

The stoichiometric coefficients of the reaction are ν_1, ν_2, ν_3, and ν_4. Cato Maximilian Gulberg and Peter Waage proposed the *law of mass action* in 1864 to describe a chemical system in equilibrium. The expression for the equilibrium constant, which defines the law of mass action, depends only on the concentration of the reactants and products at fixed temperature. To refer to concentration, we will use square brackets. For example, $[A_1]$ means the concentration of A_1. Hence, the expression of the equilibrium constant will be:

$$K_c = \frac{[\text{Products}]}{[\text{Reactants}]} = \frac{[A_3]^{\nu_3} \cdot [A_4]^{\nu_4}}{[A_1]^{\nu_1} \cdot [A_2]^{\nu_2}} \quad \text{Equilibrium constant } K_c \qquad (4.1)$$

Let us break down the expression of K_c. On top of the fraction, we have the equilibrium concentration of the products to the power of its coefficients. For example, $[A_3]^{\nu_3}$ means the equilibrium concentration of A_3 to the power of the coefficient ν_3. In the bottom of the fraction, we have the concentrations of the reactants to the power of their coefficients. All concentrations in K_c are timed.

$$K_c = \frac{\overbrace{[A_3]^{\nu_3} \cdot [A_4]^{\nu_4}}^{\text{product concentration}}}{\underbrace{[A_1]^{\nu_1} \cdot [A_2]^{\nu_2}}_{\text{reactant coefficient}}}$$

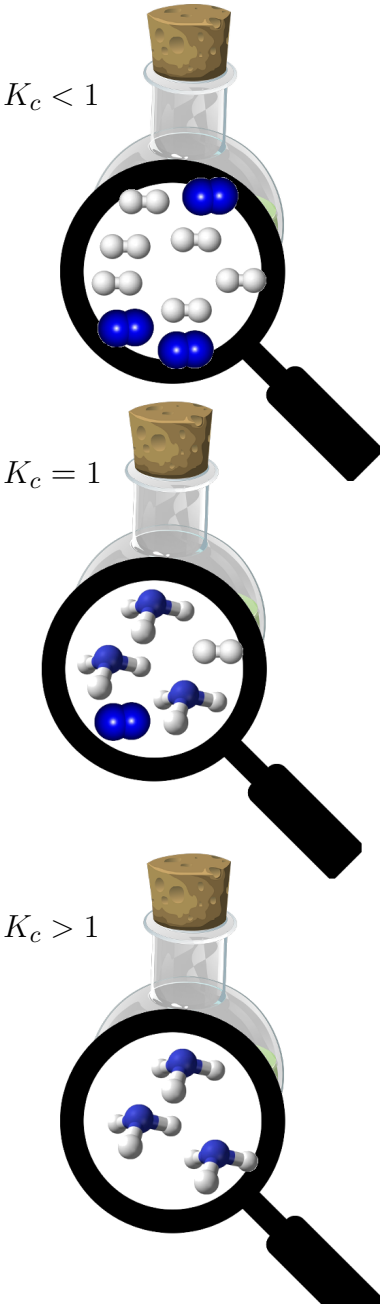
Let's focus on the reaction below:

$$C_6H_{12}O_{6(Aq)} + 6\,O_{2(Aq)} \rightleftharpoons 6\,CO_{2(Aq)} + 6\,H_2O_{(Aq)}$$

The glucose burning in the body, also known as cellular respiration. This is a controlled, multi-step biochemical process where glucose and oxygen are converted into carbon dioxide, water, as well as usable energy. The expression of the equilibrium constant for this process would be:

$$K_c = \frac{[CO_2]^6 \cdot [H_2O]^6}{[C_6H_{12}O_6] \cdot [O_2]^6}$$

As a final note, the equilibrium constant is a dimensionless property. This is because, in its more general definition, equilibrium constants are defined in terms of *chemical activities*, a general dimensionless property that relates to the amount of substance (for example, it can be concentration for solutions, pressure for gases).



▲The magnitude of the equilibrium constant indicates whether there is more products or reactants in an equilibrium mixture.

Equilibrium involving solids, liquids and solutions Let us analyze an example of a reaction involving solids or liquids. When calcium carbonate monohydrate ($CaCO_3 \cdot H_2O$) is heated, it undergoes a two-step thermal decomposition process that ultimately produces carbon dioxide gas, as shown below:

$$CaCO_3 \cdot H_2O_{(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)} + H_2O_{(l)}$$

Experiments have found that pure solids and liquids do not affect chemical equilibrium or appear in equilibrium constant expressions because their concentrations remain constant throughout a reaction. For the example above:

$$K_c = [CO_2]$$

Differently, for a reaction involving aqueous solutions, for example:
$HCl_{(Aq)} \rightleftharpoons H^+_{(Aq)} + Cl^-_{(Aq)}$ $\qquad K_c = \frac{[H^+]\cdot[Cl^-]}{[HCl]}$
Overall, the expression of K_c can only include gases (g) or aqueous solutions (aq), leaving solids and liquids out.

Sample Problem 3

Write down the expression of K_c for the following reactions:
(a) $2\,NO_{2(g)} \rightleftharpoons N_2O_{4(g)}$
(b) $C_3H_{8(g)} + 5\,O_{2(g)} \rightleftharpoons 4\,H_2O_{(g)} + 3\,CO_{2(g)}$
(c) $Zn_{(s)} + 2\,HCl_{(Aq)} \rightleftharpoons ZnCl_{2(Aq)} + H_{2(g)}$

SOLUTION

Remember you can only include gas and aqueous solution in the expression of K_c. For the first example

$$K_c = \frac{[N_2O_4]}{[NO_2]^2}$$

For the second example,

$$K_c = \frac{[H_2O]^4 \cdot [CO_2]^3}{[C_3H_8] \cdot [O_2]^5}$$

For the last example:

$$K_c = \frac{[ZnCl_2] \cdot [H_2]}{[HCl]^2}$$

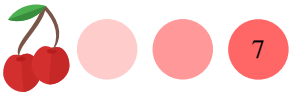
STUDY CHECK

Write down the expression of K_c for the following reaction:
$Pb^{2+}_{(Aq)} + 2\,I^-_{(Aq)} \rightleftharpoons PbI_2(s)$.

Answer: $\frac{1}{[Pb^{2+}]\cdot[I^-]^2}$

Equilibrium constant in terms of pressures The equilibrium constant K_c is the ratio of the concentrations of products and reactants. As concentration is related to pressure through the ideal gas law, we can also express an equilibrium constant in terms of pressures; we call this K_p. Let us consider, again, a general equilibrium reaction in which A_1 and A_2 react to form A_3 and A_4

$$\underbrace{\nu_1 A_{1(g)} + \nu_2 A_{2(g)}}_{\text{Reactants}} \rightleftharpoons \underbrace{\nu_3 A_{3(g)} + \nu_4 A_{4(g)}}_{\text{Products}}$$



The stoichiometric coefficients of the reaction are ν_1, ν_2, ν_3, and ν_4. Hence, the expression of the equilibrium constant in terms of pressures will be:

$$\boxed{K_p = \frac{P_{\text{Products}}}{P_{\text{Reactants}}} = \frac{P_{A_3}^{\nu_3} \cdot P_{A_4}^{\nu_4}}{P_{A_1}^{\nu_1} \cdot P_{A_2}^{\nu_2}}} \quad \text{Equilibrium constant in terms of pressures} \quad (4.2)$$

Let us break down the expression of K_p. On top of the fraction, we have the equilibrium pressures of the products to the power of their coefficients. For example, $P_{A_3}^{\nu_3}$ represents the equilibrium pressure of A_3 to the power of the coefficient ν_3. In the bottom of the fraction, we have the pressures of the reactants to the power of their coefficients. All concentrations in K_p are timed.

$$K_c = \frac{P_{A_3}^{\nu_3} \cdot P_{A_4}^{\nu_4}}{P_{A_1}^{\nu_1} \cdot P_{A_2}^{\nu_2}}$$

product pressures

reactant coefficient

Let's focus on an example. Think about the reaction below that involves just gases:

$$CH_{4(g)} + 2\,O_{2(g)} \rightleftharpoons CO_{2(g)} + 2\,H_2O_{(g)}$$

The expression of the equilibrium constant in terms of pressures would be:

$$K_p = \frac{P_{CO_2} \cdot P_{H_2O}^2}{P_{CH_4} \cdot P_{O_2}^2}$$

K_p values larger than one imply that, in the mixture, the pressure of products is larger than the pressure of reactants, and the opposite is true for values smaller than one.

`Relating` K_c `and` K_p The values of K_c and K_p are related by the following formula:

$$\boxed{K_p = K_c(RT)^{\Delta n}} \quad (4.3)$$

where:

K_p is the equilibrium constant in terms of pressure

K_c is the equilibrium constant in terms of concentration

R is the constant of the gases in pressure units (0.082 atmL/molK)

T is the absolute temperature (in Kelvins)

Δn is the change in stoichiometry of the reaction

Let us analyze the role of Δn for the combustion of propene presented below:

$$C_3H_{8(g)} + 5\,O_{2(g)} \rightleftharpoons 4\,H_2O_{(g)} + 3\,CO_{2(g)}$$

Δn represents the number of molecules of products with respect to the number of molecules of reactants:

$\Delta n = 7 - 6 = 1$, therefore for this reaction we have that $K_p = K_c(RT)$.

However, the power of the relationship will depend on the stoichiometry of the reaction. At the same time, the relationship between both constants depends on the temperature.

Sample Problem 4

For the following reaction

$$2\,NH_{3(g)} \rightleftharpoons N_{2(g)} + 3\,H_{2(g)}$$

The value of K_c at 300K is 17. Calculate the value of K_p at the same temperature.

SOLUTION

First we will write down the expression of both equilibrium constants:

$$K_c = \frac{[N_2]\cdot[H_2]^3}{[NH_3]^2} \qquad K_p = \frac{P_{N_2}\cdot P_{H_2}^3}{P_{NH_3}^2}$$

We will also calculate Δn in order to establish the relationship between both constants:

$$\Delta n = 1 + 3 - 2 = 2$$

We have that:

$$K_p = K_c(RT)^{\Delta n} = K_c(RT)^2$$

Given than K_c at 300K is 17:

$$K_p = K_c(RT)^{\Delta n} = 17(0.082\cdot 300)^2 = 10287$$

In equilibrium, the pressure of products will be larger than the pressure of reactants.

STUDY CHECK

For the following reaction

$$SO_2Cl_{2(g)} \rightleftharpoons SO_{2(g)} + Cl_{2\ (g)}$$

The value of K_p at 300K is 0.05. Calculate the value of K_c at the same temperature.

▶Answer: 2×10^{-3}

Manipulating equilibrium expressions A change in the way we write an equilibrium reaction will have an impact on the expression of the equilibrium constant, that is, on the formula representing this constant. Let us use, as an example, the *Sabatier reaction* that produces methane and water by reacting carbon dioxide with hydrogen at high temperatures and pressures over a metal catalyst

$$CO_{2(g)} + 4\,H_{2(g)} \rightleftharpoons CH_{4(g)} + 2\,H_2O_{(g)} \qquad K_c(A \rightleftharpoons B) = \frac{[CH_4]\cdot[H_2O]^2}{[CO_2]\cdot[H_2]^4}$$

The reverse of the Sabatier process is called the *steam methane reforming* (or thermal decomposition/pyrolysis of methane), which breaks down methane and water into carbon dioxide and hydrogen. The equilibrium constant is given by

$$CH_{4(g)} + 2\,H_2O_{(g)} \rightleftharpoons CO_{2(g)} + 4\,H_{2(g)} \qquad K_c(B \rightleftharpoons A) = \frac{[CO_2]\cdot[H_2]^4}{[CH_4]\cdot[H_2O]^2}$$

We can see that

$$\boxed{K_c(A \rightleftharpoons B) = \frac{1}{K_c(B \rightleftharpoons A)}}$$

Therefore, we can conclude that reversing an equilibrium inverts its equilibrium constant expression.

Similarly, if we multiply an equilibrium by a number, we also impact the expression of

the equilibrium constant. For example:

$$2\,CO_{2(g)} + 8\,H_{2(g)} \rightleftharpoons 2\,CH_{4(g)} + 4\,H_2O_{(g)} \qquad K_c(2A \rightleftharpoons 2B) = \frac{[CH_4]^2 \cdot [H_2O]^4}{[CO_2]^2 \cdot [H_2]^8}$$

By comparing $K_c(A \rightleftharpoons B)$ and $K_c(2A \rightleftharpoons 2B)$, we can see that if we time the chemical equilibrium by a number, we introduce a power on the equilibrium expression, and we have

$$\boxed{K_c(2A \rightleftharpoons 2B) = (K_c(A \rightleftharpoons B))^2}$$

Finally, let us address the case in which we add two equilibria. On one hand, we have the Sabatier reaction producing steam (reaction (1) below), but at the right temperature this steam can liquefy, producing water (reaction (2) below). The result of adding both reactions would be the Sabatier reaction producing water instead of steam (reaction (1)+(2) below)

$$\text{(1)} \quad CO_{2(g)} + 4\,H_{2(g)} \rightleftharpoons CH_{4(g)} + 2\,\cancel{H_2O_{(g)}} \qquad K_c(A \rightleftharpoons B) = \frac{[CH_4] \cdot [H_2O]^2}{[CO_2] \cdot [H_2]^4}$$

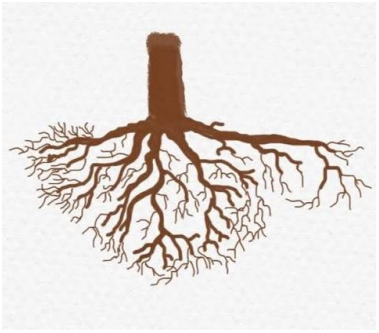
$$\text{(2)} \quad 2\,\cancel{H_2O_{(g)}} \longrightarrow 2\,H_2O_{(l)} \qquad K_c(B \rightleftharpoons C) = \frac{1}{[H_2O]^2}$$

$$\text{(1)+(2)} \quad CO_{2(g)} + 4\,H_{2(g)} \longrightarrow CH_{4(g)} + 2\,H_2O_{(l)} \qquad K_c(A \rightleftharpoons C) = \frac{[CH_4]}{[CO_2] \cdot [H_2]^4}$$

By comparing the reactions above, we can see that

$$\boxed{K_c(A \rightleftharpoons C) = K_c(A \rightleftharpoons B) \cdot K_c(B \rightleftharpoons C)}$$

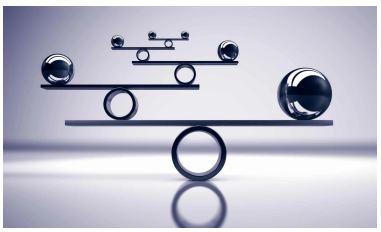
Therefore, adding two equilibria would multiply their corresponding equilibrium constants.



▲ In the 9th century, Arab writers usually called one of the equal factors of a number jadhr ('root'), and their medieval European translators used the Latin word radix



▲ Ice and water are in equilibrium in an iceberg



▲ The term equilibrium in chemistry is borrowed from physics



▲ Ammonia synthesis via the Haber-Bosch process is one of the most transformative breakthroughs in the history of chemistry

4.3 Equilibrium constants in terms of activities

Equilibrium constants can be expressed in terms of pressures as K_p or in terms of concentrations as K_c. This is because both pressure and concentration are related to the amount of matter; that is, the more concentrated the solution, the larger the amount of reactant when the reaction happens in liquid solution. Similarly, the higher the pressure, the larger the amount of reactant when the reaction happens in the gas phase, in gas mixtures. There is a general way to refer to the amount of matter called the *activity*. Activities a not only unify the different ways of expressing the amount of matter, but also take into account deviations from ideal behavior when intermolecular interactions are relevant.

The concept of activity of a chemical was coined by G. N. Lewis in 1907, and is an advanced topic covered in physical chemistry classes. By convention, activities are dimensionless quantities, defined with respect to customary choices of standard states. The activity of pure substances in condensed phases (solids and liquids) a^* is taken as one. The activity of a chemical in solution or in a gas mixture is defined as:

$$\boxed{a^*_{(l)} = 1} \quad \text{and} \quad \boxed{a^*_{(s)} = 1} \tag{4.4}$$

The definition of activity contains a coefficient called the *activity coefficient* γ, which quantifies the strength of the intermolecular forces acting between the molecules of the

chemical. The weaker this forces the close would this coeficient be to 1 and hence concentrations (or pressures) would equal to activities. But this equality only holds when the intermolecular interactions are weak.

$$a_{(aq)} = \gamma \frac{c}{c^*} \qquad a_{(g)} = \gamma \frac{p}{p^*} \tag{4.5}$$

All equilibrium constants are originally written in terms of activities, and as such are unitless magnitudes. The general definition of an equilibrium constant in terms of activities is given by

$$K = \frac{a_{\text{Products}_{(aq)}}}{a_{\text{Reactants}_{(aq)}}} \tag{4.6}$$

Using chemical activities instead of raw concentrations or partial pressures makes the equilibrium constant completely unitless and thermodynamically rigorous.

4.4 Using equilibrium constants

The equilibrium constant of a reaction indicates whether products are more abundant than reactants in a reaction mixture at equilibrium. At the same time, one can use K_c to quantitatively calculate the value of the equilibrium concentration of reactants and products, and its expression contains those variables. This section will explain how to do so.

Solving from K_c Let's analyze the reaction of hydrogen (H_2) and iodine (I_2) to produce hydrogen iodide (HI):

$$H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$$

The equilibrium constant at 300K is $3 \cdot 10^{-1}$. Analyzing a reaction mixture we find that the concentration of H_2 is 1M and the concentration of I_2 is 2M. We want to calculate how much HI we have in the mixture. As the concentrations of reactants and products are linked together through K_c we can certainly solve for $[HI]$. The expression for K_c is:

$$K_c = \frac{[HI]^2}{[H_2] \cdot [I_2]}$$

and we know that $[H_2] = 1M$ and $[I_2] = 2M$. Plugging the values in the expression of K_c, and given the numerical value of K_c we have:

$$3 \cdot 10^{-1} = \frac{[HI]^2}{1 \cdot 2}$$

Solving for $[HI]$ we have

$$[HI]^2 = 0.6$$

To solve for $[HI]$ we have to use an square root:

$$[HI] = \sqrt[2]{0.6} = 0.77M$$

Sample Problem 5

The value of the equilibrium constant in terms of concentrations for the reaction

$$3\,H_2(g) + N_2(g) \rightleftharpoons 2\,NH_3(g)$$

is $3 \cdot 10^8$ at 300K. An analysis of an equilibrium mixture gave a concentration of nitrogen and ammonia of 2M, respectively. Calculate the equilibrium concentration of hydrogen at 300K.

SOLUTION

The value of the equilibrium constant for the formation of ammonia is:

$$K_c = \frac{\left[NH_3\right]^2}{\left[H_2\right]^3 \cdot \left[N_2\right]}$$

We know $\left[NH_3\right]$ and $\left[N_2\right]$ and both values are 2M, and we also know $K_c = 3 \cdot 10^8$. Plugging these values into the previous equations we obtain:

$$3 \cdot 10^8 = \frac{2^2}{\left[H_2\right]^3 \cdot 2}$$

we can solve for $\left[H_2\right]$:

$$\left[H_2\right]^3 = 6.6 \cdot 10^{-9}$$

In order to obtain $\left[H_2\right]$ we need a a cubic root:

$$\left[H_2\right] = \sqrt[3]{6.6 \cdot 10^{-9}} = 1.9 \cdot 10^{-3} M$$

STUDY CHECK

The value of the equilibrium constant for the reaction

$$C_3H_{8(g)} + 5\,O_{2(g)} \rightleftharpoons 4\,H_2O_{(g)} + 3\,CO_{2(g)}$$

is 500 at a given temperature. An analysis of an equilibrium mixture gave a concentration of water, carbon dioxide and C_3H_8 of 1M. Calculate the equilibrium concentration of oxygen at that temperature.

Answer: $\left[O_2\right] = 0.29M$

Equilibrium constant in product form Let's consider a general equilibrium reaction in which A_1 and A_2 react to form A_3 and A_4

$$\underbrace{\nu_1 A_{1(Aq)} + \nu_2 A_{2(Aq)}}_{\text{Reactants}} \rightleftharpoons \underbrace{\nu_3 A_{3(Aq)} + \nu_4 A_{4(Aq)}}_{\text{Products}}$$

The stoichiometric coefficients of the reaction are ν_1, ν_2, ν_3, and ν_4. Mathematically speaking, K_c is represented as a fraction in which reactants appear at the bottom, whereas products appear at the top. For example, K_c for this reaction would be given by

$$K_c = \frac{\left[A_3\right]^{\nu_3} \cdot \left[A_4\right]^{\nu_4}}{\left[A_1\right]^{\nu_1} \cdot \left[A_2\right]^{\nu_2}}$$

There is an alternative, more compact, way of formulating K as a product of concentrations instead of a fraction. This can be done because we know that moving a term with

▼ Here you can learn how to carry out algebra with lists:

List Algebra with Casio fx-9750GII

To calculate for example List1∧ List2

1. **Step one:** Enter the data: start the calculator and press [menu] and the key [2] to access the list menu (stat). You will follow all the steps on this menu. Type the List 1 and List 2 values.
2. **Step two:** In menu 1, press the [OPTION] key and then the List option (key [F1]). Now press the key [1] to then pressing the power key [∧] and again the [OPTION] key and List now 2.
3. **Step three:** Press [EXE] and a List will appear populated with the result of List1∧List2 done row by row.
4. **Step four:** You can save this as List 3 by pressing [⟶] and [OPTION] List 3

▼ Here you can learn how to add the elements of a list:

List addition with Casio fx-9750GII

1. **Step one:** Press [menu] and the key [2] to access the list menu (stat). Type the List 1 values.
2. **Step two:** Press [menu] and the key [1] to access the run menu.
3. **Step two:** Press [OPTION] and [F1] for the List menu. In this menu, you will see the [▷] key that leads to [Sum], the sum function. Press it.
4. **Step three:** Now press [OPTION] and select List1.
5. **Step four:** Press [EXE] and you will obtain the result of adding all elements of List1.

▼ Here you can learn how to multiply the elements of a list:

List product with Casio fx-9750GII

1. **Step one:** Press [menu] and the key [2] to access the list menu (stat). Type the List 1 values.
2. **Step two:** Press [menu] and the key [1] to access the run menu.
3. **Step two:** Press [OPTION] and [F1] for the List menu. In this menu, you will see the [▷] key that leads to [Prod], the sum function. Press it.
4. **Step three:** Now press [OPTION] and select List1.
5. **Step four:** Press [EXE] and you will obtain the result of multiplying all elements of List1.

an exponent from the denominator to the numerator (or vice versa) changes the sign of its exponent. Using this property, we have

$$K_c = \frac{[A_3]^{\nu_3} \cdot [A_4]^{\nu_4}}{[A_1]^{\nu_1} \cdot [A_2]^{\nu_2}} = [A_1]^{-\nu_1} \cdot [A_2]^{-\nu_2} \cdot [A_3]^{\nu_3} \cdot [A_4]^{\nu_4}$$

The expression above demonstrates that ultimately, K_c results from the product of a set of different concentrations elevated to the power of their stoichiometric coefficients, which are positive for products and negative for reactants. We can hence reformulate the K_c and K_p expression s

$$\boxed{K_c = \prod_{i=1}^{4} [A_i]^{\nu_i}} \quad \text{and} \quad \boxed{K_p = \prod_{i=1}^{4} P_i^{\nu_i}} \tag{4.7}$$

This new formulation for K allows us to easily compute equilibrium constants with a graphing calculator, taking advantage of the list feature and the Prod() function. The list feature (or Stat List Editor) on a graphing calculator is a spreadsheet-like data tool that lets you enter, store, and manipulate columns of raw numerical data to perform statistical analysis, regressions, and coordinate plotting. The Prod() function calculates the product of the elements of a list.

=

Sample Problem 6

Octane (C_8H_{18}) is a hydrocarbon used as one of the standard values in the octane rating scale used to rate car fuels. Its combustion in car engines follows the equilibrium

$$2\,C_8H_{18(g)} + 25\,O_{2(g)} \rightleftharpoons 16\,CO_{2(g)} + 18\,H_2O_{(g)}$$

Compute the value of K_p given the set of equilibrium pressures below.

	$2\,C_8H_{18(g)}$	+	$25\,O_{2(g)}$	$\rightleftharpoons$	$16\,CO_{2(g)}$	+	$18\,H_2O_{(g)}$
P_i(atm)	3×10^{-2}		2×10^{-3}		2×10^{-2}		5×10^{-2}

SOLUTION

We will use the List feature and the Prod function found in graphing calculators in order to calculate K_p. First, the definition of K_p is

$$K_c = \prod_{i=1}^{4} P_i^{\nu_i} = P_1^{\nu_1} \cdot P_2^{\nu_2} \cdot P_3^{\nu_3} \cdot P_4^{\nu_4}$$

Now we will arrange the pressure values and stoichiometric coefficients, using negative values for reactants, in a table below using columns

Chemical	P_i(atm) (L1)	ν_i (L2)
$C_8H_{18(g)}$	3×10^{-2}	-2
$O_{2(g)}$	2×10^{-3}	-25
$CO_{2(g)}$	2×10^{-2}	16
$H_2O_{(g)}$	5×10^{-2}	18

We have called L1 the list containing pressures and L2 the one containing coefficients. Now, to compute K_p, we will have to do

$$K_p = \text{Prod(L1}^{\wedge}\text{L2)} = 8.3 \times 10^{19}$$

STUDY CHECK

Compute the value of K_p given the set of equilibrium pressures below.

	$3\,A_{(g)}$	+	$4\,B_{(g)}$	$\rightleftharpoons$	$13\,C_{(g)}$	+	$5\,D_{(g)}$
P_i(atm)	5×10^{-5}		3×10^{-6}		7×10^{-4}		7×10^{-3}

▶Answer: 1.6×10^{-17} .

4.5 Concentration or pressure ratio

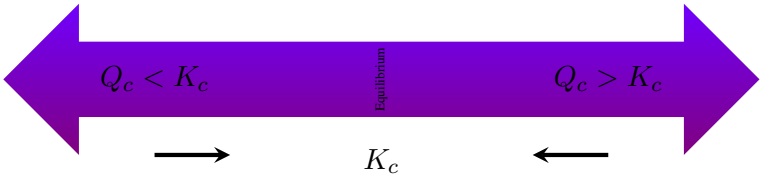
At equilibrium reactants and products exist at a very specific ratio given by the equilibrium constant. What if we would prepare a mixture containing both reactants and products? Given that both reactants and products are related by an equilibrium, how would the concentrations change? In other words, would the reaction proceed toward the right, hence producing products, or the left to produce reactants? This section introduces the use of a concentration (or pressure) ratio that helps predict the direction in which a mixture of reactants and products will proceed toward equilibrium.

Definition of concentration ratio Equilibrium constants are ratios of equilibrium concentration or pressure of products over reactants measured after the system has reacted for an infinite time. Concentration ratios Q_c are the ratio of the concentration of products over reactants away from equilibrium. In other words, they represent the reaction away from equilibrium.

$$Q_c = \frac{[\text{Products}]_{noneq}}{[\text{Reactants}]_{noneq}} = \frac{[C]^c_{noneq} \cdot [D]^d_{noneq}}{[A]^a_{noneq} \cdot [B]^b_{noneq}} \quad \text{concentration ratio} \qquad (4.8)$$

Throughout this chapter, we will use $[A]_{noneq}$ to indicate the concentration of a chemical species A away from equilibrium and $[A]$ as the concentration in equilibrium.

Use of concentration ratios Concentration ratios help predict whether a reaction will proceed toward the left or toward the right when we start with a mixture of reactants and products away from equilibrium.



If Q_c is larger than K_c the reaction will proceed towards the left producing reactants.

▼ Here you can learn how to carry out algebra with lists:

List Algebra with Texas TI-84

To calculate for example List1∧ List2

1. **Step one:** Press Stat and Enter. Type the List 1(L1) and List 2(L2) values.
2. **Step two:** Press 2nd and mode to quit from the Stat menu.
3. **Step three:** Press 2nd and 1 to operate with L1, then ∧ and then Press 2nd and 2
4. **Step four:** Press Enter and you will obtain a list resulting from multplying L1 and L2.

▼ Here you can learn how to add the elements of a list:

List addition with Texas TI-84

1. **Step one:** Press Stat and Enter. Type the List 1(L1) values.
2. **Step two:** Press 2nd and mode to quit from the Stat menu.
3. **Step three:** Press 2nd and 0 to access the catalog.
4. **Step four:** Look for the Sum() function. You can press the letter S or use the arrows up and down to locate the function. When you find the function, press Enter.
5. **Step five:** Press 2nd and 1 to operate with L1. Make sure to close the parenthesis.
6. **Step six:** Press Enter and you will obtain the result of adding all values of L1.

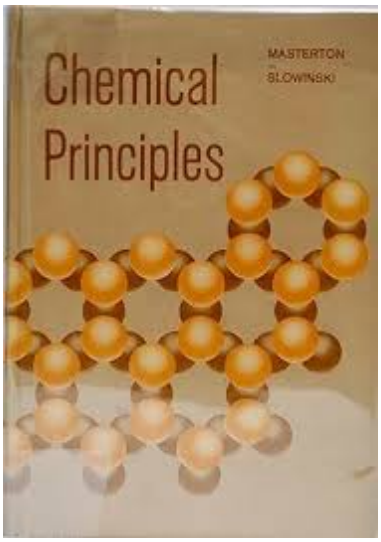
▼ Here you can learn how to multiply the elements of a list:

List product with Texas TI-84

1. **Step one:** Press Stat and Enter. Type the List 1(L1) values.
2. **Step two:** Press 2nd and mode to quit from the Stat menu.
3. **Step three:** Press 2nd and 0 to access the catalog.
4. **Step four:** Look for the Prod() function. You can press the letter S or use the arrows up and down to locate the function. When you find the function, press Enter.
5. **Step five:** Press 2nd and 1 to operate with L1. Make sure to close the parenthesis.
6. **Step six:** Press Enter and you will obtain the result of multiplying all values of L1.



▲Henry Louis Le Châtelier was a french chemist who devised a principle used by chemists to predict the effect a changing condition has on a system in chemical equilibrium.



▲An ICE table (Initial, Change, Equilibrium) dates back at least to the 1973 textbook Chemical Principles by William L. Masterton and Emil J. Slowinski, and became a standard pedagogical tool in general chemistry by the 1920s–1950s.

On the other hand, if Q_c is smaller than K_c the reaction will proceed towards the right producing products. If Q_c equals K_c then the reaction is in equilibrium.

Sample Problem 7

In two different experiments, we prepare mixtures of four chemicals (molarities indicated in the table) which are related by the following equilibrium:

$$CO_{(g)} + H_2O_{(g)} \rightleftharpoons CO_{2(g)} + H_{2(g)} \qquad K_c = 0.48$$

Indicate if any of the mixtures is in equilibrium. If it is not, indicate whether the reaction will evolve towards the left or the right to reach equilibrium.

Experiment	$CO_{(g)}$	+	$H_2O_{(g)}$	$\rightleftharpoons$	$CO_{2(g)}$	+	$H_{2(g)}$
1	0.0203		0.0203		0.0040		0.0040
2	0.011		0.0011		0.037		0.046

SOLUTION

We will compute the concentration ratio for each experiment and compare the value with k_c. If Q_c is larger than k_c the reaction will proceed to the left whereas if Q_c is smaller than k_c the reaction will proceed to the right.

Experiment	$CO_{(g)}$	+	$H_2O_{(g)}$	$\rightleftharpoons$	$CO_{2(g)}$	+	$H_{2(g)}$	Q_c
1	0.0203		0.0203		0.0040		0.0040	3.8×10^{-4}
2	0.011		0.0011		0.037		0.046	140

None of the mixtures are in equilibrium as Q_c differs from k_c. In the first experiment the reaction will proceed to the right and in the second experiment the reaction will proceed to the left.

STUDY CHECK

We prepare mixtures of four chemicals (molarities indicated in the table) in an experiment. These chemicals are related by the following equilibrium:

$$CO_{(g)} + H_2O_{(g)} \rightleftharpoons CO_{2(g)} + H_{2(g)} \qquad K_c = 0.48$$

Indicate if the mixture is in equilibrium. If it is not, indicate whether the reaction will evolve towards the left or the right to reach equilibrium.

Experiment	$CO_{(g)}$	+	$H_2O_{(g)}$	$\rightleftharpoons$	$CO_{2(g)}$	+	$H_{2(g)}$
1	0.0094		0.0025		0.0005		0.0076

▶Answer: to the right

4.6 Le Chatelier principle

At this point, we have covered the idea of equilibrium and we have seen that the forward and reverse reactions have the same rate at equilibrium. Now, what happens if you alter

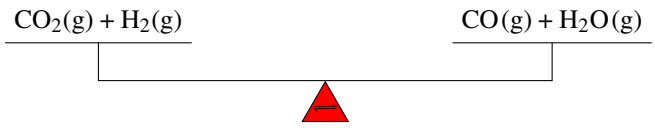
this equilibrium? Le Châtelier principle claims a reaction will go back to its original equilibrium state by shifting left or right.

Le Chatelier principle When a reaction is in equilibrium the forward and reverse reactions proceed at the same speed. Also in an equilibrium state, the concentrations of reactants and products have very specific values. Imagine that you create stress conditions by adding reactants or products or even changing the temperature. This stress will have an impact on the equilibrium and the reaction eventually will reach a new state of equilibrium by somehow counteracting this stress. Le Châtelier principle states that when stress is placed in a reaction (adding or removing reactant or products, increasing or decreasing temperature) the equilibrium will be shifted in the direction that relieves that stress. Table 4.2 displays different aspects regarding Le Châtelier's principle in terms of parameter change and consequence.

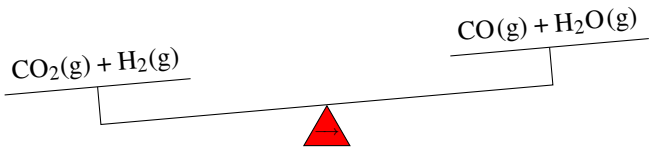
Change in concentration Let us consider the following equilibrium:

$$CO_2(g) + H_2(g) \rightleftharpoons CO(g) + H_2O(g)$$

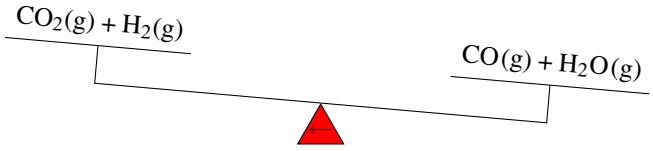
K_c for this reaction equals one at 1200K. This means that the concentration of reactants and products are the same. We can represent this using this balance or seesaw



If we add some CO_2 the equilibrium will be affected. To counteract this stress, the reaction will restore the equilibrium by decreasing the amount of CO_2. This can only be achieved by displacing the equilibrium to the right so that CO_2 is removed. Mind that CO_2 is consumed if the reaction moves from reactants $\longrightarrow$ to products and it is produced when going from products $\longleftarrow$ to reactants. We can represent this with the following seesaw.

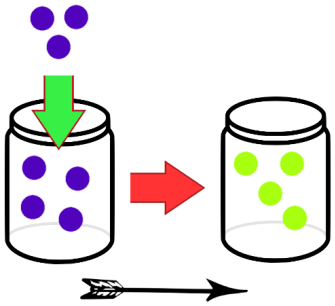


As we added CO_2 the reactants now weigh more and hence the reaction has to proceed to the right $\longrightarrow$. Now imagine we remove some $CO_2(g)$. Again, the equilibrium will be affected and the reaction will restore its equilibrium state by doing the opposite, that is producing $CO_2(g)$ as the reaction proceeds from reactants $\longleftarrow$ to products. Again using the seesaw:

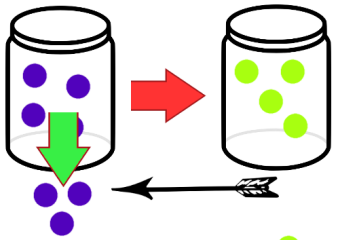


We can also add a different chemical that is not involved in the equilibrium. In this case, the equilibrium will not be affected by this change.

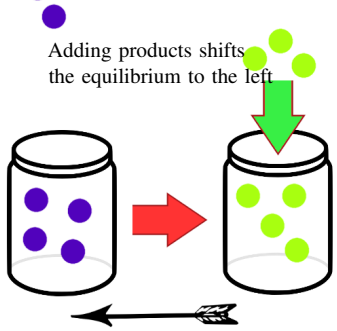
Adding reactants shifts the equilibrium to the right



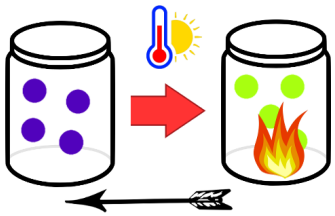
Removing reactants shifts the equilibrium to the left



Adding products shifts the equilibrium to the left



Warming an exothermic reaction shifts the equilibrium to the left

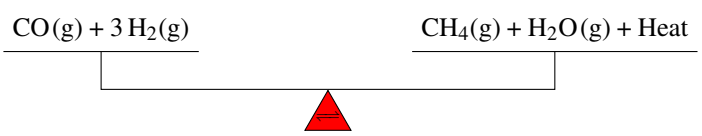


▲Le Châtelier principle helps predicts the outcome of altering an equilibrium mixture.

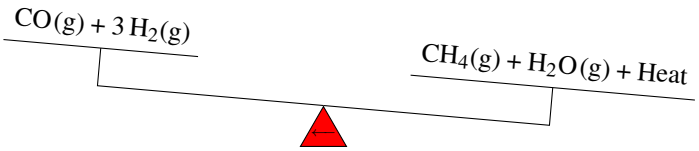
Temperature change Let us consider the following equilibrium that produces heat– remember we describe these types of reactions as exothermic:

$$\mathrm{CO\,(g)} + 3\,\mathrm{H_2(g)} \rightleftharpoons \mathrm{CH_4(g)} + \mathrm{H_2O\,(g)} + \text{Heat}$$

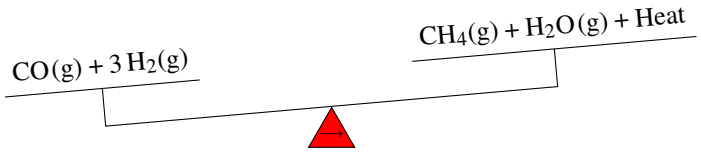
Again, this reaction is in equilibrium so we can use the same seesaw analogy.



If we increase the temperature of the system, the equilibrium will be affected. To go back to an equilibrium state the reaction will decrease the temperature of the container. As the reaction produces heat, a way to decrease the system temperature is to generate reactants (⟵). Again, using the scale that means:



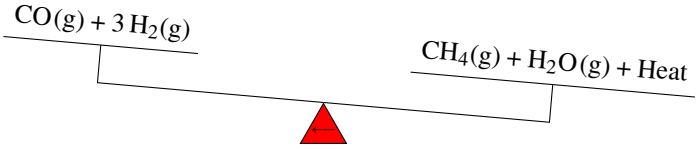
Differently now if we decrease the temperature, the reaction will increase the temperature by going back to its equilibrium state going from reactants ⟶ to products. This is because heat is produced as a product



Volume change We can also think about increasing or decreasing the volume in which the reaction takes place. This change will have an impact on the reaction equilibrium as the concentrations of reactants and products will be altered by this change. Changes in volume will shift the reaction towards the left or right depending on the overall stoichiometric change of the reaction, that is on whether the reaction produces or consumes molecules. For reactions that generate matter, that is, in the case that $\Delta n > 0$, increasing the volume will follow the increase of the number of moles. In other words, by increasing the volume, the equilibrium will shift towards the products, that is towards the right. For reactions that consume matter ($\Delta n < 0$), increasing the volume will shift the equilibrium towards the reactants, that is towards the left. For example, the reaction below consumes molecules:

$$\mathrm{CO\,(g)} + 3\,\mathrm{H_2(g)} \rightleftharpoons \mathrm{CH_4(g)} + \mathrm{H_2O\,(g)}$$

if we increase the volume of the container in which the reaction takes place, the equilibrium will shift toward the left:





▲ Jacobus van't Hoff Jr. was a Dutch physical chemist who won the first-ever Nobel Prize in Chemistry in 1901



▲ High-pressure chemistry studies chemical reactions and material transformations conducted at pressures significantly above atmospheric levels



▲ A chemical reactor for gas-phase reactions



▲ High-temperature chemistry studies chemical and physical phenomena occurring above 500 K (440 °F / 227 °C)

The opposite shift will follow a volume decrease as the reaction shift towards the right.

Table 4.2 Le Châtelier principle

Parameter	Change	Effect
Concentration	Add reactants	$\longrightarrow$
	Remove reactants	$\longleftarrow$
	Add products	$\longleftarrow$
	Remove products	$\longrightarrow$
Volume	Increase volume	More moles
	Decrease volume	Less moles
Catalyst	Add a catalyst	No effect
Temperature		
Endothermic reaction	Increase temperature	$\longrightarrow$
	Decrease temperature	$\longleftarrow$
Exothermic reaction	Increase temperature	$\longleftarrow$
	Decrease temperature	$\longrightarrow$



▲A visible spectrophotometer determines a chemical equilibrium constant by measuring how much light a colored species absorbs in solution, which directly reveals its concentration via the Beer-Lambert Law ($A = \kappa \cdot c$).



▲ Fritz Haber and Robert Le Rossignol (1907–1908): Accurately measured the ammonia equilibrium under pressure (30-200 atm), proving that high pressures could force the yield of ammonia to commercially viable levels at temperatures around 500-600°C using metal catalysts like osmium or uranium

Sample Problem 8

For the next endothermic reaction indicate whether the reaction will shift right ($\longrightarrow$) or left ($\longleftarrow$) after the following changes:

$$N_{2(g)} + O_{2(g)} + \text{Heat} \rightleftharpoons 2\,NO_{(g)}$$

(a) adding reactants (b) adding products (c) decreasing the temperature.

SOLUTION

(a) Adding reactants always displaces the equilirbium so that reactants are consumed, hence the reaction will proceed $\longrightarrow$. (b) After adding products the reaction will tend to reduce the amount of products, and hence it will go $\longleftarrow$. (c) The reaction is endothermic that means that it consumes heat. If we decrease the temperature it will tend to increase the temperature and hence heat needs to be formed. This will only happen if the reaction proceeds ($\longleftarrow$).

STUDY CHECK

For next exothermic reaction indicate whether the reaction will shift right ($\longrightarrow$) or left ($\longleftarrow$) after the following changes:

$$C_{(g)} + O_{2(g)} \rightleftharpoons CO_{2(g)} + \text{Heat}$$

(a) removing reactants (b) removing products (c) decreasing the temperature.

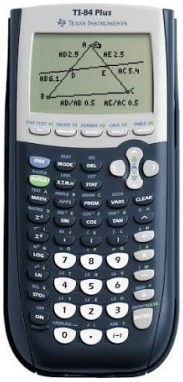
▶Answer: (a) $\longleftarrow$; (b) $\longrightarrow$; (c) $\longrightarrow$.

4.7 ICE tabular method

The ICE method is a tabular method used to obtain the extent of a reaction. This method leads to a mathematical function, a polynomial. Its roots give the numerical value of the



▲ A casio graphing calculator



▲ A texas graphing calculator

extent of a reaction, allowing the calculation of the equilibrium concentration of all species involved in the chemical equilibrium. Experimentally, we can obtain the equilibrium concentration of the chemical species involved in an equilibrium using, for example, absorption spectroscopy. Using spectroscopy, one can measure the amount of product formed, for example, and hence elucidate the amount of reactants decomposed. The ICE methods help us calculate those concentrations based on the principles of chemical equilibrium.

The ICE method The ICE method is a tabular method that helps compute the extent of a reaction, represented by the unknown variable x, for a given equilibrium. It requires the filling of a table, called the ICE table. Each column of this table represents a reactant or a product, whereas the rows represent the different stages of a reaction reaching equilibrium: initial, change, and equilibrium, with the first row displaying the chemical reaction whose equilibrium we are interested in studying. All quantities in the table are expressed in amount of matter units (for example, molar or atmosphere units). Remember that both molarity and gas pressure inform about the amount of matter and can be unified in a more general property called the *activity*.

There are a few rules to follow when filling the ICE table. First, the values of x, the extent of the reaction, should respect the reaction's stoichiometry and, for example, if the stoichiometric coefficient of a reactant is 2, the corresponding entry on the table for that reactant should be $2x$. Second, the entry should include negative signs for reactants and positive signs for the products. This is because reactants decompose, disappear to produce products. Also, the state of matter does matter, and solids or liquids should not be included in the table. The last, perhaps more important, row is the equilibrium, E, row. It contains the result of adding the I and C rows.

Below we have examples of ICE charts for the reaction between carbon monoxide and water to produce carbon dioxide, with initial concentrations of 0.1M and 0.2 M, respectively, for carbon monoxide and water.

	$CO_{(g)}$	+	$H_2O_{(g)}$	$\rightleftharpoons$	$CO_{2(g)}$	+	$H_{2(g)}$
Inicial	0.1M		0.2M		0M		0M
Change	$-x$		$-x$		x		x
Equilibrium	$0.1-x$		$0.2-x$		x		x

Another example for for the decomposition of nitrosyl chloride ($NOCl_{(g)}$)–used primarily as a chemical intermediate, a reagent in organic synthesis, and historically for flour bleaching–at an initial concentration of 0.3M to produce nitrogen monoxide and chlorine gas

	$2\,NOCl_{(g)}$	$\rightleftharpoons$	$2NO_{(g)}$	+	$Cl_{2(g)}$
Inicial	0.3M		0M		0M
Change	$-2x$		$2x$		x
Equilibrium	$0.3-2x$		$2x$		x

Extent of a reaction The extent of a reaction, named x–and often ξ–is the extent to which the reaction proceeds until reaching equilibrium. This concept, introduced around 100 years ago by belgian scientist Théophile de Donder, is also known as the progress of a reaction, advancement of the reaction, or the conversion. Some reactions will proceed so that most of the reactants are decomposed into products, and hence the extent will be large, whereas others will proceed to a lesser extent so that

fewer reactants become products. x is a single value for a reaction depending on the equilibrium constant value, temperature and the starting concentrations of reactants and products in the mixture. When using the ICE tabular method the extent of the reaction is represented as x.

A polynomial associated with chemical equilibrium The last row of the ICE chart represents the equilibrium concentrations of reactants and products and is the most important row of the ICE table. However, these concentrations are expressed in terms of the extent of the reaction x and will remain unknown unless we know x. By plugging the equilibrium concentrations in terms of x into K_c, we can obtain a polynomial $P(x)$ in terms of x and eventually solve for x.

$$\boxed{P(x) = 0}$$

The positive root of this polynomial–or the single root that makes chemical sense–will lead to the numerical value for the extent of reaction. Roots without chemical sense are negative roots or roots larger than the initial concentration of reactants give. For example, it does no chemical sense if we start with a reactant concentration of 1M and we obtain a root ot 2M, which would mean that 2M would decompose from 1M. Similarly, negative roots would not make chemical sense.

Each reaction in equilibrium has an associated polynomial with an order that is related not only to the complexity and type of equilibrium but also to the initial concentrations of reactants and products. For example, acid-base equilibrium tends to be represented by second-order polynomials, whereas solubility equilibrium can be represented by third-degree polynomials. The use of polynomials to describe an equilibrium is not new, and this family of mathematical functions is well known in fields such as game theory, fluid mechanics, or statistical mechanics. You can use your graphing calculator to obtain the roots of any of the polynomials associated with a chemical equilibrium. Always remember only the positive root–or the root that makes chemical sense–has chemical meaning.

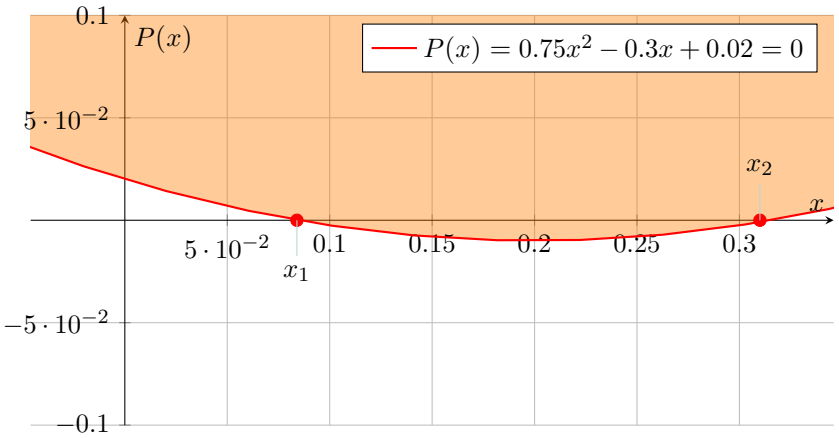


Figure 4.4 Polynomial $P(x)$ associated with the $CO_{(g)} + H_2O_{(g)} \rightleftharpoons CO_{2(g)} + H_{2(g)}$ equilibrium starting with 0.1 and 0.2M respectively for $CO_{(g)}$ and $H_2O_{(g)}$. Two roots are obtained: x_1=0.08 M and x_2=0.3 M. Only x_1=0.08 M has chemical meaning, as it is smaller than 0.1 and 0.2M.

Figure 4.4 displays $P(x)$ for the reaction between carbon monoxide and water, starting with a 0.1M concentration of CO and a 0.2 M concentration of H_2O. The equilibrium



▲ Rene Descartes (1596-1650): Introduced modern algebraic notation (using letters for variables like x) and popularized equating polynomial expressions to zero.



▲ Théophile De Donder was a Belgian mathematician, physicist, and chemist who first developed the concept of the extent of a reaction. He was in 1927, one of the participants in the fifth Solvay Conference on Physics, which took place at the International Solvay Institute for Physics in Belgium.

▼ Here you can learn how to find polynomial roots:

Polynomial roots with Casio fx-9750GII

1. **Step one:** Press Menu and 8.
2. **Step two:** Press the Poly (F2) menu. and select the degree. Now enter the coefficients of the polynomial and press *Solve*.
3. **Step three:** At this point, you will have all roots

▼ Here you can learn how to find polynomial roots using a web app:

Polynomial roots with Alpertron

1. **Step one:** Go to the website www.alpertron.com.ar/POLFACT.HTM
2. **Step two:** Type the polynomial using the symbol ^ to write powers and press factor to get the roots.
3. **Step three:** Change the sign on the factors obtained in order to obtain the roots.

▼ Here you can learn how to find polynomial roots using a texas graphing calculator:

Polynomial roots with Texas TI 84

1. **Step one:** Texas calculators do not provide a default app to calculate polynomial roots and you will have to install the PolySmlt2 external App.

constant for this reaction at 527 °C is 4. We can see that the polynomial is a parabola with two roots x_1=0.08 M and x_2=0.3 M. Only the first root would represent the extent of the reaction x as we cannot produce 0.3M from either 0.1M or 0.2M. Therefore, only x_1=0.08 M has chemical sense.

Sample Problem 9

The following polynomials are associated with chemical equilibria. For each one, obtain the extent of the reaction x, indicating the roots without chemical sense: (a) $P_1(x) = -0.9x^2 + 2.2x - 1 = 0$, starting with a initial concentration of 0.8M (b) $P_2(x) = 27x^4 - 54x^3 - 1630.7x^2 - 10x + 1 = 0$, staring with a initial concentration of 1M

SOLUTION

We can obtain the roots of the polynomials using for example a graphing calculator or using the *quadratic equation solver link*. (a) The roots for $P_1(x)$ are x_1=0.60 and x_2=1.84. The second root has no chemical sense as they say we start with 0.8M. (b) The roots for $P_2(x)$ are x_1=0.022, x_2=8.8, x_3=−0.028, and x_4=−6.83M. The last three roots have no chemical sense as they say we start with 1M and the extent can not be negative.

STUDY CHECK

The following polynomials are associated with chemical equilibriua. For each one, obtain the extent of the reaction, indicating the roots without chemical sense: (a) $P_1(x) = -221.2x^2 + 112.6x - 13.38 = 0$, starting with a initial concentration of 0.28M (b) $P_2(x) = 4x^3 + 0.8x^2 + 0.55x - 0.049 = 0$, starting with a initial concentration of 0.1M

▶Answer: (a) x_1=0.189 and x_2=0.319(no chemical sense); (b) x=0.077

4.8 Equilibrium calculations

In the previous sections, we covered how to properly build ICE charts and the fact that each chemical equilibrium has an associated polynomial. The roots of these polynomials are related to the extent x of the reaction at equilibrium. As such, it was critical to be able to obtain polynomial roots by means of a graphing calculator. Now, we will address the use of ICE charts to study reactions at equilibrium, and we will describe not only how to explicitly obtain the associated polynomial, but also how to use its roots to obtain chemical information.

Equilibrium calculations The ICE tabular method is a powerful method to obtain the equilibrium concentration of all species involved in an equilibrium. Here we lay the foundations of the steps needed to obtain those quantities. First we, set up the ICE table for the equilibrium, indicating the initial concentrations (or pressures) of the mixture to then obtain the expression of K (sometimes K_p others K_c or simply K) in terms of the extent x to then compute the polynomial associated $P(x)$. Now we obtain the roots of $P(x)$ and plug them back into the ICE chart. Finally, at this point, the last line of the table will give the equilibrium concentrations. The side Figure visally explains this method.

ConcentrationSideFigure-ammonias at equilibrium We can easily ob-

tain the equilibrium concentrations of reactants and products once we have obtained the positive root of the equilibrium polynomial, the extent of the reaction x. Let us analyze the reaction between water and carbon monoxide to produce carbon dioxide and hydrogen given that the initial concentration of water is 0.2M and the one for carbon monoxide is 0.1M

	$CO_{(g)}$	+	$H_2O_{(g)}$	$\rightleftharpoons$	$CO_{2(g)}$	+	$H_{2(g)}$
Inicial	0.1M		0.2M		0M		0M
Change	$-x$		$-x$		x		x
Equilibrium	$0.1-x$		$0.2-x$		x		x

we can plug the value of the extent of reaction to calculate the equilibrium concentrations. For the example in hand, we found that the reaction extent was x, hence we have that the equilibrium concentrations, which is K_c=5.1 at 700K, are:

$$K_c = \frac{x^2}{(0.1-x)\cdot(0.2-x)} = \frac{x^2}{x^2-0.3x+0.02}$$

rearanging we have

$$\Big(1-K_c\Big)x^2 + 0.3\cdot K_c\cdot x - 0.02\cdot K_c = 0$$

now substituying K_c we have rearanging we have

$$P(x) = -4.1x^2 + 1.53x - 0.102 = 0 \tag{4.9}$$

which is the polynomial $P(x)$ associated with the equilibrium for the given set of initial temperature and concentration conditions. Solving to x we arrive to two different solutions x_1=0.286M and x_2=0.087M. The first answer makes no chemical sense, as we start with 0.1M of CO and 0.286M cannot decompose from 0.1M. So we have that the final reaction extent is 0.086M. Now we can go back to our ICE chart and compute the equilibrium concentration of each reactant and product. The ICE chart incorporating the solved value for x would be

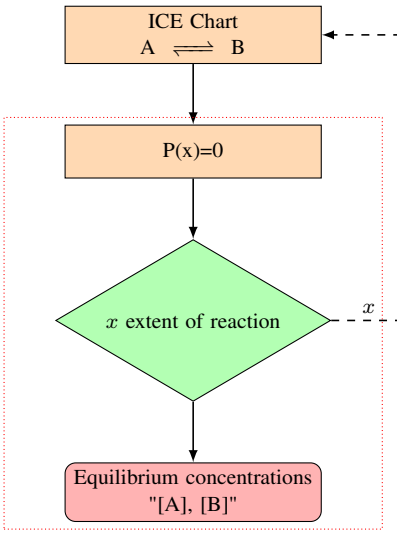
	$CO_{(g)}$	+	$H_2O_{(g)}$	$\rightleftharpoons$	$CO_{2(g)}$	+	$H_{2(g)}$
Inicial	0.1M		0.2M		0M		0M
Change	-0.086		-0.086		0.086		0.086
Equilibrium	0.013M		0.113M		0.087M		0.087M

The ICE chart above displays, on one hand, the extent of the reaction as 0.086 M of reactants decompose, and 0.086 M of products are formed. It also displays the concentration equilibrium of all chemical species involved in the equilibrium:

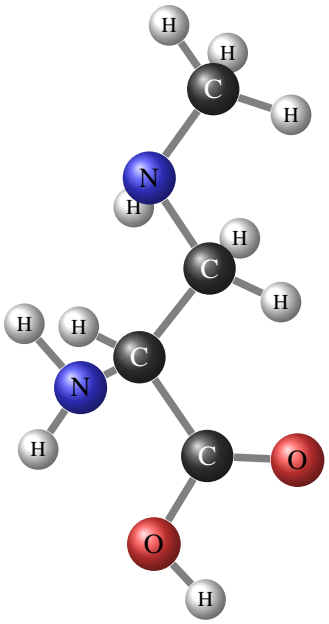
$[CO]$=0.013M, $[H_2O]$=0.113M, $[CO_2]$=0.087M and $[H_2]$=0.087M.

We can verify that using these equilibrium concentration we obtain the value of K_c:

$$K_c = \frac{[CO_2]\cdot[H_2]}{[CO]\cdot[H_2O]} = \frac{0.087\cdot 0.087}{0.013\cdot 0.113} = 5.1$$



▲ Workflow diagram displaying the use of the ICE tabular method to compute equilibrium concentrations.



▲ In water, amino acids exist in a chemical equilibrium state with its Zwitterion form

Sample Problem 10

A mixture of 2 mol of N_2 and 2 mol of H_2 was placed in a 2L vessel at 720°C. At that temperature, the equilibrium constant for the formation of ammonia is K_c=2.4×10^{-3}

$$H_{2(g)} + 3\,H_{2(g)} \rightleftharpoons 2\,NH_{3(g)}$$

Once the equilibrium is reached, calculate the equilibrium concentration of the reactants and products.

SOLUTION

We will first build up the ICE chart given that the initial concentration of reactants is both 1M

	$N_{2(g)}$ +	$3H_{2(g)}$ $\rightleftharpoons$	$2NH_{3(g)}$
Initial	1M	1M	0M
Change	$-x$	$-3x$	$2x$
Equilibrium	$1-x$	$1-3x$	$2x$

Now we will obtain the polynomial and solve for the extent. Using the expression for K_c

$$K_c = \frac{[ceNH3]^2}{[N_2]\cdot[H_2]^3}$$

now substituting the values from the ICE chart:

$$K_c = \frac{(2x)^2}{(1-x)\cdot(1-3x)^3}$$

By opening parenthesis we obtain

$$K_c = \frac{4x^2}{27x^4 - 54x^3 + 36x^2 - 10x + 1}$$

The corresponding polynomial will be

$$27x^4 - 54x^3 + \left(36 - \frac{4}{K_c}\right)x^2 - 10x + 1 = 0$$

The final polynomial will be

$$P(x) = 27x^4 - 54x^3 - 1630.7x^2 - 10x + 1 = 0$$

By solving given that K_c=2.4×10^{-3}, we obtain four different roots, and only the positive root lower than one, $x = 0.022$ has chemical sense. Mind that x can not be smaller than the given initial concentration of 1M. The other roots are: 8.8, -0.028, and -6.83M. Now we can obtain the equilibrium concentrations for reactants and products: $[N_2]$=0.978M, $[H_2]$=0.934M, and $[NH_3]$=0.044M based on the ICE chart

	$N_{2(g)}$ +	$3H_{2(g)}$ $\rightleftharpoons$	$2NH_{3(g)}$
Initial	1M	1M	0M
Change	-0.022	-0.066	0.044
Equilibrium	$0.978M$	$0.934M$	$0.044M$

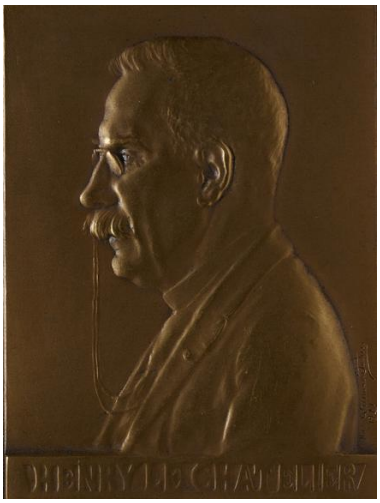
STUDY CHECK

Hydrogen reacts with molecular iodine in gaseous state to produce hydroiodic acid, a strong, colorless, corrosive gas with acidic properties. Suppose that a mixture made of 0.1moles of H_2, 0.3moles of I_2, and 0.01moles of HI is placed in a 0.5L flask. Given the equilibrium constant for the reaction at a given temperature

$$H_{2(g)} + I_{2(g)} \rightleftharpoons 2\,HI_{(g)} \qquad K_c = 100$$

calculate the equilibrium concentration of all species.

▶Answer: 0.035, 0.435, and 0.35M.



▲ The Le Chatelier Medal is the highest distinction in cement and concrete research, awarded annually by the international scientific journal Cement and Concrete Research

General polynomial expressions We know the ICE tabular method is used to obtain the equilibrium concentrations of all species involved in an equilibrium. The steps behind the method involve building a table and computing the roots of a polynomial associated with the equilibrium. This method is, in general, tedious, as the specific steps involving the rearranging of the equilibrium constant in terms of x can be complex based on the powers involved in K_c.

We can simplify this process by applying the ICE tabular method in a more general way, so that we can obtain general expressions for the polynomials associated with different equilibrium scenarios. Once we have these general formulas, which apply to all equilibria with the same stoichiometric coefficients, we can simply apply them to specific scenarios, leading to final numerical expressions for the equilibrium polynomials.

The first equilibrium we will address involves the reaction A + B $\longrightarrow$ C + D. The following ICE chart represents the equilibrium between four chemical scpecies: A, B, C and D.

	$A_{(g)}$	+	$B_{(g)}$	$\rightleftharpoons$	$C_{(g)}$	+	$D_{(g)}$
Inicial	c_a		c_b		c_c		c_d
Change	$-x$		$-x$		x		x
Equilibrium	$c_a - x$		$c_b - x$		$c_c + x$		$c_d + x$

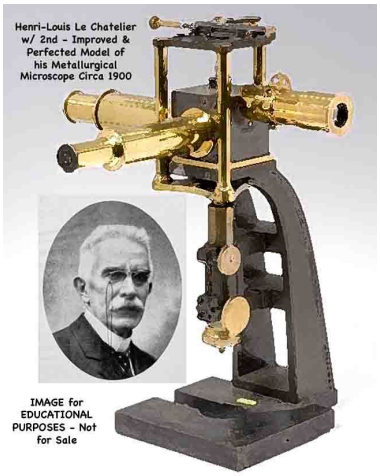
Now we will obtain the polynomial and solve for the extent. Using the expression for K_c

$$K_c = \frac{[C]\cdot[D]}{[A]\cdot[B]} = \frac{(c_c + x)\cdot(c_d + x)}{(c_a - x)\cdot(c_b - x)} = \frac{x^2 + (c_c + c_d)x + (c_c \cdot c_d)}{x^2 - (c_a + c_b)x + (c_a \cdot c_b)}$$

Now, rearranging we have a general expression for the polynomial associated to the equilibrium:

$$\boxed{\Big[1 - K_c\Big]x^2 + \Big[K_c \cdot (c_a + c_b) + (c_c + c_d)\Big]x + \Big[c_c \cdot c_d - K_c \cdot c_a \cdot c_b\Big] = 0} \tag{4.10}$$

The second equilibrium we will address involves the reaction A + B $\longrightarrow$ 2 C. The following ICE chart represents the equilibrium between four chemical scpecies: A, B, and C.



▲ Le Chatelier designed an inverted metallographic microscope in the late 19th century to examine heavy and large metallic and minera



▲ Le Chatelier co-authored the seminal scientific book about gaseous explosions

	$A_{(g)}$	+	$B_{(g)}$	$\rightleftharpoons$	$2\,C_{(g)}$
Inicial	c_a		c_b		c_c
Change	$-x$		$-x$		$2x$
Equilibrium	$c_a - x$		$c_b - x$		$c_c + 2x$

Now we will obtain the polynomial and solve for the extent. Using the expression for K_c

$$K_c = \frac{[\mathrm{C}]^2}{[\mathrm{A}]\cdot[\mathrm{B}]} = \frac{(c_c + 2x)^2}{(c_a - x)\cdot(c_b - x)} = \frac{4x^2 + (4\cdot c_c)x + c_c^2}{x^2 - (c_a + c_b)x + (c_a \cdot c_b)}$$

Now, rearranging we have a general expression for the polynomial associated to the equilibrium:

$$\boxed{\Big[4 - K_c\Big]x^2 + \Big[K_c\cdot(c_a + c_b) + 4c_c\Big]x + \Big[c_c^2 - K_c\cdot c_a\cdot c_b\Big] = 0} \tag{4.11}$$

We can now put to use these general expressions for P(x) by addressing the following equilibrium, given that K_c is 5.1 at 700K

	$CO_{(g)}$	+	$H_2O_{(g)}$	$\rightleftharpoons$	$CO_{2(g)}$	+	$H_{2(g)}$
Inicial	0.1M		0.2M		0M		0M
Change	$-x$		$-x$		x		x
Equilibrium	$0.1 - x$		$0.2 - x$		x		x

Using Equation 4.10, substituying 0.1M in c_a, 0.2M in c_b, and 0M in c_c and c_c, as well as 5.1 in K_c, we directly obtain

$$P(x) = -4.1x^2 + 1.53x - 0.102 = 0$$

which is equivalent to Equation 4.9.

Equilibrium calculations involving gases For equilibrium systems involving gases, we can use the measurement of the total gas pressure in equilibrium to experimentally measure the extent of the reaction x, and hence not only to measure how the reaction advances towards equilibrium, but also to experimentally measure the equilibrium constant.

We will use the decomposition of gaseous nitrosyl chloride $NOCl_{(g)}$ as an example. Starting with an initial 0.50atm pressure of NOCl and heating up the system at 500K, we have the ICE chart below in terms of pressures.

	$2\,NOCl_{(g)}$	$\rightleftharpoons$	$2NO_{(g)}$	+	$Cl_{2(g)}$
Inicial	0.50atm		0atm		0atm
Change	$-2x$		$2x$		x
Equilibrium	$0.50 - 2x$		$2x$		x

When carrying out this experiment using a pressure sensor, we notice that the total pressure of the reaction increases from 0.50atm to 0.537atm, with a total increase of 0.037atm. The goal is to measure the extent x of the reaction and to calculate its equilibrium constant in terms of pressures, K_p.

The total initial pressure would be P_i=0.50atm, as indicated on the ICE chart, whereas the total pressure at equilibrium would result from adding the partial pressures of all gases

$$P_e = P_{\mathrm{NOCl}} + P_{\mathrm{NO}} + P_{\mathrm{Cl_2}} = (2x) + (x) + (0.5 - 2x) = x + 0.50\mathrm{atm}.$$

We can see that the value of x corresponds to the pressure increase of the reaction: 0.037atm. We can now compute K_p using the formula

$$K_p = \frac{P_{\text{NO}}^2 \cdot P_{\text{Cl}_2}}{P_{\text{NOCl}}^2} = \frac{(2x)^2 \cdot (x)}{(0.5-2x)^2} = \frac{4x^3}{(0.5-2x)^2} = \frac{4 \cdot 0.037^3}{(0.5-2\cdot 0.037)^2} = 1.1 \times 10^{-3}$$

In summary, when dealing with gases, the experimental measurement of the pressure in equilibrium can be used to compute reaction extents and equilibrium constants.

Percent dissociation Once the extent of the reaction x is known we can also obtain the percent dissociation of the reactants α which is just the amount that has been dissociated with the respect to the initial quantities placed on the reaction container in percent form. Each reactants can have a different percent dissociation but it all deppends on the reaction stoichiometry and the initial quantities. The general formula for this property is

$$\boxed{\%\alpha_a = \frac{\nu_a \cdot x}{c_a} \times 100}$$

where:

α_a is the percent dissociation of the reactant a

ν_a is the stiochiomtric coficiente of the reactant a in the equilibrium

c_a is the initial concentration of the reactant a

For example, for the between carbon monoxide and water presented in the following ICE chart, we can compute the percent dissociations for CO and H_2O

	$CO_{(g)}$	+	$H_2O_{(g)}$	$\rightleftharpoons$	$CO_{2(g)}$	+	$H_{2(g)}$
Inicial	0.1M		0.2M		0M		0M
Change	-0.086M		-0.086M		0.086M		0.086M
Equilibrium	0.013M		0.113M		0.087M		0.087M

For the case of CO we have that its stoichiometric coeficient is 1 and its initial concentration is 0.1M. Given that the extent of the reaction is 0.086 we have

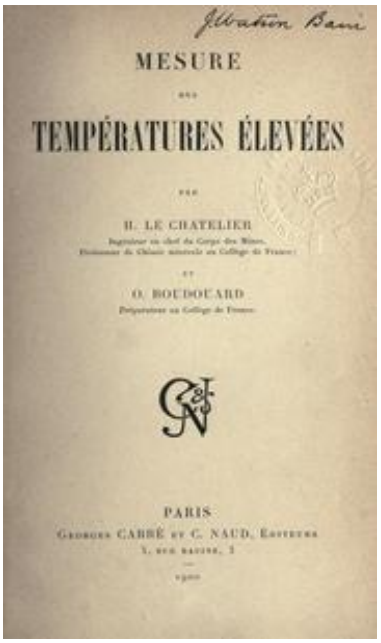
$$\%\alpha_{\text{CO}} = \frac{\nu_{\text{CO}} \cdot x}{c_{\text{CO}}} \times 100 = \frac{1 \cdot 0.086}{0.1} \times 100 = 86\%$$

For the case of H_2O, the percent dissociation was 43%. We can even incorporate the percent dissociation values into the ICE chart

	$CO_{(g)}$	+	$H_2O_{(g)}$	$\rightleftharpoons$	$CO_{2(g)}$	+	$H_{2(g)}$
Inicial	0.1M		0.2M		0M		0M
Change	-0.086M		-0.086M		0.086M		0.086M
	(-86%)		(-43%)				
Equilibrium	0.013M		0.113M		0.087M		0.087M

we can see that carbon monoxide dissociates more than water.

Reactions with small equilibrium constant Some of the equilibrium constant values are small, which means the amount of reactants decomposing into products x is also small. That scenario simplifies, to a large extent, the degree of the polynomials involved in the equilibria. For example, the equilibrium constant for the formation of ammonia at 720°C is K_c=2.4×10^{-3}



▲ Le Chatelier's book about experimental methods for measuring extreme heat.



▲ Synthetic ammonia production is directly tied to global food security, with synthetic nitrogen fertilizers currently sustaining nearly half of the world's population and preventing mass starvation



▲ Ammonia played a critical role in the Second World War as a foundational chemical for manufacturing explosives, gunpowders, and synthetic nitrates required for military ordnance

$$H_{2(g)} + 3\,H_{2(g)} \rightleftharpoons 2\,NH_{3(g)}$$

being this value small. An ICE chart for this reaction would be

	$N_{2(g)}$ +	$3H_{2(g)}$ $\rightleftharpoons$	$2NH_{3(g)}$
Initial	1M	1M	0M
Change	$-x$	$-3x$	$2x$
Equilibrium	$1-x$	$1-3x$	$2x$

As K_c is small, we expect x to be small too, and therefore we can simplify the ICE table given that

$$1 - x \simeq 1 \quad \text{and} \quad 1 - 3x \simeq 1$$

Hence the new ICE chart would be

	$N_{2(g)}$ +	$3H_{2(g)}$ $\rightleftharpoons$	$2NH_{3(g)}$
Initial	1M	1M	0M
Change	$-x$	$-3x$	$2x$
Equilibrium	~ 1	~ 1	$2x$

With this simplified ICE chart, we will obtain the polynomial and solve for the extent. Using the expression for K_c and substituting the values from the ICE chart:

$$K_c = \frac{[NH_3]^2}{[N_2]\cdot[H_2]^3} \simeq \frac{(2x)^2}{(1)\cdot(1)^3} = \frac{4x^2}{1}$$

The corresponding polynomial will be

$$P(x) \simeq 4x^2 - K_c = 0$$

This approach simplifies P(x) by reducing its degree and hence makes it easier to solve for x. The positive root would be $x \sim 0.024$M, this value is similar to the exact one calculated in the example above (0.022M). We can verify our approach given the calculated x value with the percent error indicated in parenthesis

$$1 - x = 0.97(2\%) \quad \text{and} \quad 1 - 3x = 0.92(7\%)$$

These results indicate the error is less than 10%. Similarly, we see that the approximate x value (0.024M) and the exact one (0.022M) also differ by less than 10%. In general, we can use this approach when the predicted approximate values differ by less than 10% from the exact values.



▲ The equilibrium between dinitrogen tetroxide and nitrogen dioxide is a classic reversible chemical system used to demonstrate Le Chatelier's principle.



▲ Chile saltpetre is the common name for sodium nitrate, was the world's primary natural source of agricultural fertilizer and gunpowder raw material until the invention of the synthetic Haber-Bosch process for ammonia production



▲ Carl Bosch scaled up Fritz Haber's laboratory discovery into the industrial Haber-Bosch process, enabling the mass production of synthetic ammonia from atmospheric nitrogen and hydrogen

4.9 Effect of temperature on K

The equilibrium constant of a reaction is related to the thermodynamic characteristics of the reaction. This means it depends on the difference in energy between reactants and products and not on the rate of conversion between reactants and products. The equilibrium constant of a reaction does not depend on the initial concentration or pressure of the species involved. However, K deppends on temperature and the change deppends at the same time on the energetic character of the reaction. Exothermic reaction, reactions producing heat, are favoured at low temperature, whereas endothermic reactions, reactions consuming heat, are favoured at high temperature. This section qualitatively addresses the change of K_p with temperature, introducing the van't Hoff equation.

van't Hoff equation The van't Hoff equation gives the dependency of the equilibrium constant in terms of pressure with temperature. We will introduce this equation in two forms: a linearized form involving one temperature, suitable for linear regression calculations when you have numerous temperatures, and a form involving two temperatures, suitable for punctual calculations involving only two points

$$\boxed{Ln\left[\frac{K_p(T_2)}{K_p(T_1)}\right] = \frac{\Delta H_f^\circ}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]} \text{ and } \boxed{LnK_p(T) = c - \frac{\Delta H_f^\circ}{R}\frac{1}{T}} \quad (4.12)$$

where:

K_p is the equilibrium constant in terms of pressures a temperature T

ΔH_r° is the enthalpy change of the reaction in J/mol

R is the constant of the gases in energy units (8.314J/molK)

T is the temperature in Kelvins

c is a constant



▲ Ammonia is used to produce fertilizers.



▲ Minnesota now has a wind-powered green ammonia plant.

The linear form of the equation is useful to obtain data graphically, as it represents a linear relationship with $\frac{1}{T}$ as x and $Ln(k)$ as y. The slope of the resulting line is $-\frac{\Delta H_r^\circ}{R}$ (mind the negative sign) and the intercept is c. Based on the van't Hoff equation, the larger the enthalpy, the smaller the change of the equilibrium constant with temperature. Similarly, as the temperature decreases K_p increases if the reaction is endothermic, and decreases when the reaction is exothermic. Some advice, as ΔH_r° are often given in KJ/mol, make sure you convert the value in J/mol before you use the van't Hoff equation.

The following example shows how to for this equation to a large set of temperatures.



▲ True organic fertilizers do not contain synthetic ammonia, though organic nitrogen sources (like blood meal, fish emulsion, or plant meals) naturally release ammonium and nitrates into the soil as microbes break them down.

Sample Problem 11

Using the following data and a regression analysis, calculate the the standard enthalpy of reaction for the synthesis of ammonia. Is the reaction endothermic or exothermic.

$N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3$				
T (K)	298	320	350	370
K_p	6.80×10^5	5.29×10^4	2.73×10^3	4.94×10^2

SOLUTION

As they provide with rate constant and temperature data, we will use the linearized form of Arrhenius equation in order to calculate the activation energy and the frequency factor. We will first process the data and compute the natural logarithm of the rate constant and the inverse of temperature. By analyzing the data we can see that the equilibrium constant decreases as temeprature rises and therefore the reaction has to be exothermic



▼ You need to follow the following steps to carry out a linear regression in a Casio fx 9750GII:

Linear Regresion using Casio fx 9750GII

1. **Step one:** Enter the data: start the calculator and press menu+2 (stat) to access the list menu. You will follow all the steps on this menu. Type the X values in List 1 and Y in List 2.
2. **Step two:** Press Calc (F2) on the bottom menu of the screen, then press Reg (F3) to press X (F1) and then $ax + b$ (F1).
3. **Step three:** At this point, you will have all the results from the regression on the screen.

▼ You need to follow the following steps to carry out a linear regression in a TI 84:

Linear Regresion using Texas TI 84

1. **Step one:** Enter the data: press Stat and Enter. Type the X values in List 1 and Y in List 2.
2. **Step two:** Press Second and Zero to access the catalog. Now, go down the menu until you find DiagnosticOn and select it.
3. **Step three:** Now carry out the regression by pressing Stat and Calc. Now go down on the menu until you find $ax + b$ and press it.
4. **Step four:** At this point, you will have all the results from the regression on the screen.

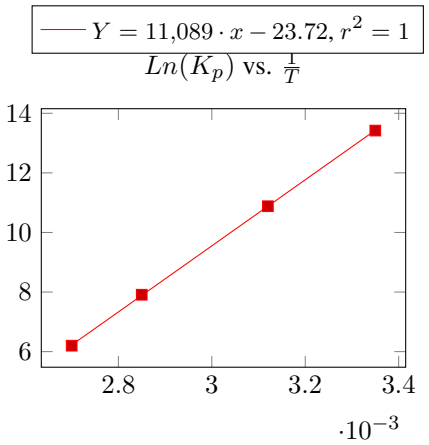
▼ You need to follow the following steps to carry out a linear regression using the webApp GraphPad:

Linear Regresion using GraphPad

1. **Step one:** Go to www.graphpad.com/quickcalcs/linear1
2. **Step two:** Enter the data into the table and click on *Calculate Now* to obtain the results
3. **Step three:** You will find the slope listed and the r^2 listed under the Goodness of fit section
4. **Step four:** the error in the slope is also listed

T (K)	$\frac{1}{T}$	K_p	$Ln(K_p)$
298	3.35×10^{-5}	6.80×10^{5}	13.43
320	3.12×10^{-3}	5.29×10^{4}	10.88
350	2.86×10^{-3}	2.73×10^{3}	7.91
370	2.70×10^{-3}	4.94×10^{2}	6.20

Now we are ready to graph the van't Hoff plot with $\frac{1}{T}$ in the horizontal (x) axis and $Ln(K_p)$ in the vertical (y) axis.



From the slope of the graph we will calculate $-\frac{\Delta H_r^\circ}{R}$ and from the intersect we will calculate c. We have that $-\frac{\Delta H_r^\circ}{R} = +11088$ and $c = -23.70$. We have that the enthalpy is: $\Delta H_r^\circ = -92.2 KJ/mol$ and the reaction is exothermic.

STUDY CHECK

Using the following data, calculate the enthalpy of reaction for the following gas-phase reaction using a regression analysis:

$N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3$			
T (K)	290	400	500
K_p	1.50×10^{-1}	$4.79 \times 10^{+2}$	$1.70 \times 10^{+3}$

▶Answer: $\Delta H_r^\circ = 57.11 KJ/mol$.

CHAPTER 4

CHEMICAL EQUILIBRIUM

4.1 Indicate if the following statement is true or false: (a) At equilibrium, the rate of the reverser reaction is twice the rate of the forward reaction (b) At equilibrium, the concentration of products do not change with time (c) At equilibrium, the concentration of reactants do not change with time (d) At equilibrium, the concentration of reactants and products do not change with time

4.2 Indicate if the following statement is true or false: (a) At equilibrium, the rate of the reverse reaction do not change (b) At equilibrium, the rate of the forward reaction do not change (c) At equilibrium, the rate of the reverse reaction equals the rate of the forward reaction (d) At equilibrium, the concentration of reactants and products are not constant

EQUILIBRIUM CONSTANTS

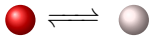
4.3 Write down the forward and reverse reactions for the following reactions in equilibrium: (a) $CH_{4(g)} + O_{2(g)} \rightleftharpoons CO_{2(g)} + H_2O_{(g)}$ (b) $2\,Mg_{(s)} + O_{2(g)} \rightleftharpoons 2\,MgO_{(s)}$

4.4 Write down the forward and reverse reactions for the following reactions in equilibrium: (a) $BaCO_{3(s)} \rightleftharpoons Ba^{2+}_{(aq)} + {CO_3}^{2-}{}_{(aq)}$ (b) $2\,H_{2(g)} + O_{2(g)} \rightleftharpoons 2\,H_2O_{(l)}$

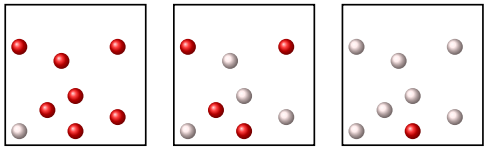
4.5 For the reactions below and given the value of the equilibrium constant indicate whether the equilibrium mixture will have: (a) More reactants than products (b) More products than reactants (c) Same amount of products and reactants (a) $CO_{2(g)} + 2\,H_2O_{(g)} \rightleftharpoons CH_{4(g)} + 2\,O_{2(g)}$ $K_c = 0.001$ (b) $N_{2(g)} + O_{2(g)} \rightleftharpoons 2\,NO_{(g)}$ $K_c = 2 \cdot 10^{25}$ (c) $2\,NO_{(g)} + O_{2(g)} \rightleftharpoons 2\,NO_{2(g)}$ $K_c = 6.4 \cdot 10^{9}$

4.6 For the reactions below and given the value of the equilibrium constant indicate whether the equilibrium mixture will have: (a) More reactants than products (b) More products than reactants (c) Same amount of products and reactants (a) $N_{2(g)} + 3\,H_{2(g)} \rightleftharpoons 2\,NH_{3(g)}$ $K_c = 1$ (b) $2\,NO_{(g)} + Cl_{2(g)} \rightleftharpoons 2\,NOCl_{(g)}$ $K_c = 6.5 \cdot 10^{4}$

4.7 Indicate which of the following diagrams (left, center, right) represent better the following reaction at equilibrium given the K_c values (a, b, c).

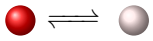


For example, you might say the "left diagram corresponds to for example (a)".

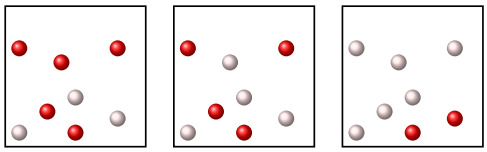


(a) $K_c = 7$ (b) $K_c = 0.1$ (c) $K_c = 1$

4.8 Indicate which of the following diagrams (left, center, right) represent better the following reactiion at equilibrium given the K_c values (a, b, c).



For example, you might say the "left diagram corresponds to for example (a)".



(a) $K_c = 1$ (b) $K_c = 0.6$ (c) $K_c = 3$

4.9 Write down the expressions of K_c and K_p for the following reactions: (a) $2\,SO_{2(g)} + O_{2(g)} \rightleftharpoons 2\,SO_{3(g)}$ (b) $CO_{(g)} + 2\,H_{2(g)} \rightleftharpoons CH_3OH_{(g)}$ (c) $C_2H_{6(g)} + Cl_{2(g)} \rightleftharpoons C_2H_2Cl_{(s)} + HCl_{(g)}$

4.10 Write down the expression of K_c for the following reactions: (a) $BaCO_{3(s)} \rightleftharpoons Ba^{2+}_{(aq)} + {CO_3}^{2-}{}_{(aq)}$ (b) $2\,H_{2(g)} + O_{2(g)} \rightleftharpoons 2\,H_2O_{(l)}$

4.11 For the reaction below at a given temperature

$$2\,SO_{2(g)} + O_{2(g)} \rightleftharpoons 2\,SO_{3(g)} \qquad K_c = 4 \times 10^{24}$$

What is the K_c value for the following reactions at the same temperature? (a) $4\,SO_{2(g)} + 2\,O_{2(g)} \rightleftharpoons 4\,SO_{3(g)}$ (b) $SO_{2(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons SO_{3(g)}$ (c) $6\,SO_{2(g)} + 3\,O_{2(g)} \rightleftharpoons 6\,SO_{3(g)}$

4.12 For the reaction below at a given temperature

$$2\,BrCl_{(g)} \rightleftharpoons Br_{2(g)} + Cl_{2(g)} \qquad K_c = 377$$

What is the K_c value for the following reactions at the same temperature? (a) $4\,BrCl_{(g)} \rightleftharpoons 2\,Br_{2(g)} + 2\,Cl_{2(g)}$ (b) $2\,Br_{2(g)} + 2\,Cl_{2(g)} \rightleftharpoons 4\,BrCl_{(g)}$ (c) $6\,BrCl_{(g)} \rightleftharpoons 3\,Br_{2(g)} + 3\,Cl_{2(g)}$

Using equilibrium constants

4.13 The reaction of carbon monoxode with hydrogen to produce methanol has a equilibrium constant in terms of concentration that at a certain temperature is larger than one

$$CO(g) + 2\,H_2(g) \rightleftharpoons CH_3OH(g) \qquad K_c = 14.5$$

Calculate: (a) the equilibrium concentration of hydrogen (H_2) given that the equilibrium concentration of methanol (CH_3OH) and carbon monoxide (CO) for the reaction is 2M, respectively. (b) the equilibrium concentration of hydrogen (H_2) given that the equilibrium concentration of methanol (CH_3OH) and carbon monoxide (CO) for the reaction are 3M and 1M, respectively.

4.14 Consider the following reaction:

$$2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g)$$

(a) Write down the expression of K_c. (b) Calculate the numerical value of K_c for the reaction if the concentrations at equilibrium at 1000K are 2M for SO_3, 0.3M for O_2 and 1M for SO_2. (c) indicate whether an equilibrium mixture will contain mostly products, mostly reactants or equal amounts of reactants and products.

4.15 Complete the table and calculate K_c and K_p at 300K:

	Reaction	K_c	K_p
(a)	$2\,NH_{3(g)} \rightleftharpoons N_{2(g)} + 3\,H_{2(g)}$	17	
(b)	$SO_2Cl_{2(g)} \rightleftharpoons SO_{2(g)} + Cl_{2(g)}$		0.05

4.16 Complete the table and calculate K_c and K_p at 300K:

	Reaction	K_c	K_p
(a)	$2\,SO_{3(g)} \rightleftharpoons 2\,SO_{2(g)} + O_{2(g)}$	0.243	
(b)	$Cl_{2(g)} + Br_{2(g)} \rightleftharpoons 2\,BrCl_{2(g)}$		0.196

Concentration ratio

4.17 For the reactions below indicate whether they will evolve towards the right or towards the left in order to reach equilibrium.

	Reaction	K_c	Q
(a)	$2\,NH_{3(g)} \rightleftharpoons N_2 + 3\,H_2$	17	20
(b)	$2\,SO_{3(g)} \rightleftharpoons 2\,SO_{2(g)} + O_{2(g)}$	0.243	10

4.18 For the reactions below indicate whether they will evolve towards the right or towards the left in order to reach equilibrium.

	Reaction	K_c	Q
(a)	$H_{2(g)} + I_2 \rightleftharpoons 2\,HI_{(g)}$	50	0.1
(b)	$H_2O_{(l)} \rightleftharpoons H_2O_{(g)}$	0.196	0.196

4.19 For the decomposition of calcium chloride hexahydrate

$$CaCl_2 \cdot 6\,H_2O(s) \rightleftharpoons CaCl_2(s) + 6\,H_2O(g)$$

we have that K_c=3.5 $\times$ 10^{-54} and Q=10 at 300K. Indicate towards which direction the reaction will evolve to reach equilibrium.

4.20 For the decomposition of calcium chloride hexahydrate

$$2\,NH_{3(g)} \rightleftharpoons N_2 + 3\,H_2$$

we have that K_c=17 and Q=10 at 300K. Indicate towards which direction the reaction will evolve to reach equilibrium.

4.21 In two different experiments (each one presented in a row from the table below), we prepare mixtures of four chemicals (molarities indicated in the table) interconnected by the following equilibrium:

$$CO_{(g)} + H_2O_{(g)} \rightleftharpoons CO_{2(g)} + H_{2(g)} \qquad K_c = 0.48$$

Indicate if any of the mixtures is in equilibrium. If it is not, indicate whether the reaction will evolve towards reactants or products to reach equilibrium.



$CO_{(g)}$	+	$H_2O_{(g)}$	$\rightleftharpoons$	$CO_{2(g)}$	+	$H_{2(g)}$
0.01		0.02		0.05		0.001
0.20		0.50		0.60		0.10

4.22 In two different experiments (each one presented in a row from the table below), we prepare mixtures of four chemicals (molarities indicated in the table) interconnected by the following equilibrium:

$$CO_{(g)} + H_2O_{(g)} \rightleftharpoons CO_{2(g)} + H_{2(g)} \qquad K_c = 0.48$$

Indicate if any of the mixtures is in equilibrium. If it is not, indicate whether the reaction will evolve towards reactants or products to reach equilibrium.

$CO_{(g)}$	+	$H_2O_{(g)}$	$\rightleftharpoons$	$CO_{2(g)}$	+	$H_{2(g)}$
0.04		0.02		0.02		0.04
0.05		0.10		0.80		0.001

Le Châtelier principle

4.23 Using the Le Châtelier principle indicate whether the reaction below

$$2\,SO_2(g) + O_2(g) \rightleftharpoons 2\,SO_3(g)$$

will shift in the direction of products ($\longrightarrow$) or reactants ($\longleftarrow$) after the following actions: (a) add SO_2 (b) add SO_3(c) remove O_2

4.24 Using the Le Châtelier principle indicate whether the reaction below

$$2\,NH_{3(g)} \rightleftharpoons N_2 + 3\,H_2$$

will shift in the direction of products ($\longrightarrow$) or reactants ($\longleftarrow$) after the following actions: (a) add NH_3 (b) add N_2(c) remove H_2

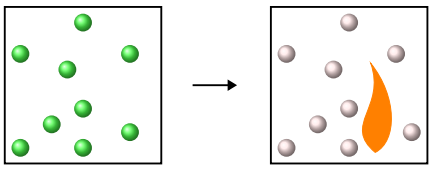
4.25 According to Le Châtelier principle indicate whether the reaction will shift in the direction of products ($\longrightarrow$) or reactants ($\longleftarrow$) after we increase temperature:

$$CO(g) + 2\,H_2(g) \rightleftharpoons CH_3OH(g) + \text{Heat}$$

4.26 According to Le Châtelier principle indicate whether the reaction will shift in the direction of products ($\longrightarrow$) or reactants ($\longleftarrow$) after we increase temperature:

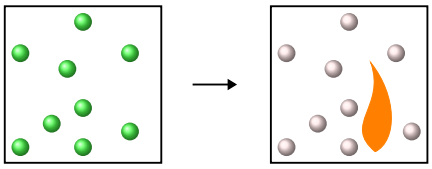
$$2\,B(s) + 3\,H_2(g) + \text{Heat} \rightleftharpoons B_2H_6(g)$$

4.27 According to Le Châtelier principle indicate whether the following reaction will shift in the direction of products ($\longrightarrow$) or reactants ($\longleftarrow$) after the following changes:



(a) adding reactants (b) increasing temperature (c) decreasing temperature

4.28 According to Le Châtelier principle indicate whether the following reaction will shift in the direction of products ($\longrightarrow$) or reactants ($\longleftarrow$) after the following changes:



(a) adding products (b) removing products (c) increasing temperature

The ICE tabular method

4.29 At 298K, sulfur dioxide can be oxidized to produce sulfur trioxide according to the following reaction

$$2\,SO_{2(g)} + O_{2(g)} \rightleftharpoons 2\,SO_{3(g)} \qquad K_c = 4 \times 10^{24}$$

Obtain the ICE chart for the following scenarios: (a) A sample of 5.0 mmol $SO_{2(g)}$ was sealed into a 2.0-L reaction vessel and heated with 1.0 mmol $O_{2(g)}$ (b) A sample of 1.0 mol $SO_{2(g)}$ was sealed into a 2.0-L reaction vessel and heated with 5 mol $O_{2(g)}$ and 5 mol $SO_{3(g)}$ (c) A sample of 4.0 mol $SO_{2(g)}$ was sealed into a 2.0-L reaction vessel and heated with 1 mol $SO_{3(g)}$

4.30 Bromine monochloride (BrCl) is primarily used as an industrial biocide, algaecide, fungicide, and disinfectant for wastewater and recirculating cooling water systems. At 300K, bromine monochloride dissociates to produce gaseous bromine and chloride.

$2\,BrCl_{(g)} \rightleftharpoons Br_{2(g)} + Cl_{2(g)} \qquad K_c = 377$

Obtain the ICE chart for the following scenarios: (a) A sample of 5 atm of $BrCl_{(g)}$ and 1 atm of $Br_{2(g)}$ was sealed into a 2.0-L reaction vessel. (b) A sample of 1 atm of $BrCl_{(g)}$ and, 1 atm of $Br_{2(g)}$ 5 atm of $Cl_{2(g)}$ was sealed into a 4.0-L reaction vessel. (c) A sample of 2 atm of $Br_{2(g)}$ and 1 atm of $Cl_{2(g)}$ was sealed into a 4.0-L reaction vessel.

4.31 For the equilibrium stablished between the chemical species A, B, C and D

$$A_{(g)} + 2\,B_{(g)} \rightleftharpoons 3\,C_{(g)} + 2\,D_{(g)}$$

assess the validity of the associated ICE chart with given inicial concentrations. Correct the ICE chart in the case it is not properly set up.

	$A_{(g)}$ +	$2B_{(g)}$ $\rightleftharpoons$	$3C_{(g)}$ +	$2D_{(g)}$
Initial	1M	1M	1M	1M
Change	$-x$	$-2x$	$+3x$	$+2x$
Equilibrium	$1-x$	$1-2x$	$1+3x$	$1+2x$

4.32 For the equilibrium stablished between the chemical species A, B, C and D

$$2\,A_{(g)} + B_{(g)} \rightleftharpoons 4\,C_{(g)} + 2\,D_{(g)}$$

assess the validity of the associated ICE chart with given inicial concentrations. Correct the ICE chart in the case it is not properly set up.

	$2A_{(g)}$ +	$B_{(g)}$ $\rightleftharpoons$	$4C_{(g)}$ +	$2D_{(g)}$
Initial	1M	1M	1M	1M
Change	$-2x$	$-x$	$+4x$	$+2x$
Equilibrium	$1-2x$	$1-x$	$1+x$	$1+2x$

4.33 For the equilibrium stablished between the chemical species A, B, C and D

$$3\,A_{(g)} + B_{(s)} \rightleftharpoons 5\,C_{(g)} + 2\,D_{(g)}$$

assess the validity of the associated ICE chart with given inicial concentrations. Correct the ICE chart in the case it is not properly set up.

	$3A_{(g)}$ +	$B_{(s)}$ $\rightleftharpoons$	$5C_{(g)}$ +	$2D_{(g)}$
Initial	1M	1M	1M	1M
Change	$-3x$	$-x$	$+5x$	$+2x$
Equilibrium	$1-3x$	$1-x$	$1+5x$	$1+2x$

4.34 For the equilibrium stablished between the chemical species A, B, C and D

$$2\,A_{(s)} + 1\,B_{(g)} \rightleftharpoons 9\,C_{(g)} + 1\,D_{(g)}$$

assess the validity of the associated ICE chart with given inicial concentrations. Correct the ICE chart in the case it is not properly set up.

	$2A_{(s)}$ +	$B_{(g)}$ $\rightleftharpoons$	$9C_{(g)}$ +	$1D_{(g)}$
Initial		1M	1M	1M
Change		$-x$	$+9x$	$+1x$
Equilibrium		$1-x$	$1+9x$	$1+1x$

4.35 The following polynomials are associated with chemical equilibria. For each one, obtain the extent of the reaction x, indicating the roots without chemical sense: (a) $P_1(x) = -4x^3 + 1.2x^2 + 0.09x + 0.002 = 0$, starting with a initial concentrations of 0.1M and 0.2M (b) $P_2(x) = 4x^2 + 0.208x - 1.5 \times 10^{-3} = 0$, starting with a initial concentrations of 0.5M and 0.05M

4.36 The following polynomials are associated with chemical equilibria. For each one, obtain the extent of the reaction x, indicating the roots without chemical sense: (a) $P_1(x) = 4x^2 + 0.048x - 7 \times 10^{-4} = 0$, starting with a initial concentrations of 0.1M and 0.01M (b) $P_2(x) = -4x^3 + 0.24x^2 - 4.5 \times 10^{-1}x + 2.5 \times 10^{-5} = 0$, starting with a initial concentrations of 0.01M and 0.05M

Equilibrium calculations

4.37 For the dissociation of iodine

$$I_{2(g)} \rightleftharpoons 2\,I_{(g)}$$

obtain the polynomial associated with the following equilibrium given the set of initial concentration and temperature conditions: (a) A sample of 5 moles of $I_{2(g)}$ and 1×10^{-2} moles of $I_{(g)}$ was sealed into a 1.0-L reaction vessel at 800K given than at that temperature $K_c = 3.1 \times 10^{-5}$. (b) A sample of 1 moles of $I_{2(g)}$ and 1×10^{-3} moles of $I_{(g)}$ was sealed into a 1.0-L reaction vessel at 800K given than at that temperature $K_c = 3.1 \times 10^{-5}$. (c) A sample of 3 moles of $I_{2(g)}$ and 1×10^{-3} moles of $I_{(g)}$ was sealed into a 1.0-L reaction vessel at 800K given than at that temperature $K_c = 3.1 \times 10^{-5}$. (d) A sample of 3 moles of $I_{2(g)}$ was sealed into a 1.0-L reaction vessel at 800K given than at that temperature $K_c = 3.1 \times 10^{-5}$. (e) A sample of 3 moles of $I_{2(g)}$ was sealed into a 1.0-L reaction vessel at 1200K given than at that temperature $K_c = 6.8 \times 10^{-2}$.



4.38 For the dissociation of fluorine,

$$F_{2(g)} \rightleftharpoons 2\,F_{(g)}$$

obtain the polynomial associated with the following equilibrium given the set of initial concentration and temperature conditions: (a) A sample of 1 moles of $F_{2(g)}$ and 1×10^{-3} moles of $F_{(g)}$ was sealed into a 1.0-L reaction vessel at 1200K given than at that temperature $K_c = 2.7 \times 10^{-3}$. (b) A sample of 1 moles of $F_{2(g)}$ was sealed into a 1.0-L reaction vessel at 1200K given than at that temperature $K_c = 2.7 \times 10^{-3}$. (c) A sample of 1×10^{-3} moles of $F_{(g)}$ was sealed into a 1.0-L reaction vessel at 1200K given than at that temperature $K_c = 2.7 \times 10^{-3}$. (d) A sample of 1×10^{-3} moles of $F_{(g)}$ was sealed into a 1.0-L reaction vessel at 1000K given than at that temperature $K_c = 1.2 \times 10^{-4}$. (e) A sample of 1 moles of $F_{2(g)}$ and 5×10^{-3} moles of $F_{(g)}$ was sealed into a 1.0-L reaction vessel at 1000K given than at that temperature $K_c = 1.2 \times 10^{-4}$.

4.39 The reaction between carbon monoxide and water vapor is known as the Water-gas shift reaction, which produces carbon dioxide and hydrogen gas. This process provides a less expensive, more efficient method of hydrogen production, serving as a hydrogen source for industrial processes such as the Haber-Bosch ammonia synthesis. The equilibrium constant at 700K is

$$CO_{(g)} + H_2O_{(g)} \rightleftharpoons CO_{2(g)} + H_{2(g)} \qquad K_c = 10$$

A mixture consisting of 1 mol $H_2O_{(g)}$ and 1 mol $CO_{(g)}$ is placed in a 1-L reaction vessel at 700. K. At equilibrium, 1 mol $CO_{2(g)}$ is present. Calculate (a) the polynomial associated to the equilibrium at those initial concentration and temperature conditions (b) the extent x of the reaction. (c) Is there a root of the polynomial that makes no chemical sense? Why? (d) the equilibrium concentrations for all substances

4.40 The reaction between carbon monoxide and water vapor is known as the Water-gas shift reaction, which produces carbon dioxide and hydrogen gas. This process provides a less expensive, more efficient method of hydrogen production, serving as a hydrogen source for industrial processes such as the Haber-Bosch ammonia synthesis. The equilibrium constant at 700K is

$$CO_{(g)} + H_2O_{(g)} \rightleftharpoons CO_{2(g)} + H_{2(g)} \qquad K = 10$$

A mixture consisting of 1 mol $H_2O_{(g)}$, 1 mol $CO_{(g)}$ and 2 mol $CO_{2(g)}$ is placed in a 1-L reaction vessel at 700. K. At equilibrium, 1 mol $CO_{2(g)}$ is present. Calculate (a) the polynomial associated with the equilibrium at those initial concentration and temperature conditions (b) the extent x of the reaction. (c) is there a root of the polynomial that makes no sense chemically speaking? Why? (d) the equilibrium concentrations for all substances

4.41 At 700 K, hydrogen iodide decomposes to hydrogen and iodide following the reaction

$$2\,HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)} \qquad K = 0.018$$

If 1.0 mol HI, 2.0 mol H_2 and 1.0 mol I_2 are placed into a 2.0-L flask, calculate (a) the polynomial associated with the equilibrium at those initial concentration and temperature conditions (b) the extent x of the reaction. (c) Is there a root of the polynomial that makes no chemical sense? Why? (d) the equilibrium concentrations for all substances

4.42 At 700 K, hydrogen iodide decomposes to hydrogen and iodide following the reaction

$$2\,HI_{(g)} \rightleftharpoons H_{2(g)} + I_{2(g)} \qquad K = 0.018$$

If 1.0 mol HI, and 1.0 mol H_2 are placed into a 2.0-L flask, calculate (a) the polynomial associated with the equilibrium at those initial concentration and temperature conditions (b) the extent x of the reaction. (c) Is there a root of the polynomial that makes no chemical sense? Why? (d) the equilibrium concentrations for all substances

4.43 Hydrogen deuteride (HD) is an isotopologue of dihydrogen (H_2) composed of two isotopes of hydrogen: the majority isotope H-1 (protium, H) and H-2 (deuterium, D). It is primarily used as an astrophysical tracer to study cold cosmic gas, the primary fuel for star formation in the universe. A 5-L flask was filled with 3 mol of gaseous HD is heated. Assume that the reaction

$$2\,HD_{(g)} \rightleftharpoons H_{2(g)} + D_{2(g)} \qquad K = 0.26$$

occurs under these conditions. Calculate (a) the ICE chart in molarity units (b) the polynomial associated with the equilibrium at those initial concentration and temperature conditions (c) the extent x of the reaction. (d) the equilibrium concentrations for all substances

4.44 Hydrogen deuteride (HD) is an isotopologue of dihydrogen (H_2) composed of two isotopes of hydrogen: the majority isotope H-1 (protium, H) and H-2 (deuterium, D). It is primarily used as an astrophysical tracer to study cold cosmic gas, the primary fuel for star formation in the universe. A 3-L flask was filled with 1 mol of gaseous HD and 1 mol gaseous H_2 is heated. Assume that the reaction

$$2\,HD_{(g)} \rightleftharpoons H_{2(g)} + D_{2(g)} \qquad K = 0.26$$

occurs under these conditions. Calculate (a) the ICE chart (b) the polynomial associated with the equilibrium at those initial concentration and temperature conditions (c) the extent x of the reaction. (d) the equilibrium concentrations for all substances

4.45 At a particular temperature, 10 moles of $SO_{2(g)}$ are mixed with 12 moles of $O_{2(g)}$ to form $SO_{3(g)}$ in a 1L container according to

$$2\,SO_{2(g)} + O_{2(g)} \rightleftharpoons 2\,SO_{3(g)}$$

At equilibrium, 2 mol of $SO_{2(g)}$ are present. Calculate (a) the ICE chart in terms of the extent of the reaction x (b) the extent x of the reaction (c) the equilibrium concentrations for all substances (d) the equilibrium constant

4.46 At a particular temperature, 5 moles of $SO_{2(g)}$ are mixed with 5 moles of $O_{2(g)}$ and 5 moles of $SO_{3(g)}$ in a 2L container. These react according to

$$2\,SO_{2(g)} + O_{2(g)} \rightleftharpoons 2\,SO_{3(g)}$$

At equilibrium, 1 mol of $SO_{2(g)}$ are present. Calculate (a) the ICE chart in terms of the extent of the reaction x (b) the extent x of the reaction (c) the equilibrium concentrations for all substances (d) the equilibrium constant

4.47 The equilibrium constant for the reaction at 500K is indicated below

$$2\,SO_{2(g)} + O_{2(g)} \rightleftharpoons 2\,SO_{3(g)} \qquad K_c = 2.5 \times 10^{10}$$

An initial mixture of sulfur dioxide and oxygen reacts in a rigid container. At equilibrium, the concentrations are $\left[SO_{2(g)}\right]_e = 0.1M$, $\left[O_{2(g)}\right]_e = 1.5M$, $\left[SO_{3(g)}\right]_e = 1M$. Answer the following questions: (a) using a and b as the concentrations of sulfur dioxide and oxygen gas that were reacted initially, obtain the ICE chart in terms of the extent of the reaction x (b) obtain an expression for the polynomial associated with the equilibrium (c) obtain the extent of the reaction (d) what were the concentrations of sulfur dioxide and oxygen gas that were reacted initially? (e) obtain an expression for the polynomial associated with the equilibrium

4.48 The equilibrium constant for the reaction at 500K is indicated below

$$2\,SO_{2(g)} + O_{2(g)} \rightleftharpoons 2\,SO_{3(g)} \qquad K_c = 2.5 \times 10^{10}$$

An initial mixture of sulfur dioxide, oxygen, and sulfur trioxide reacts in a rigid container. The concentration of oxygen in this mixture is 1M. At equilibrium, the concentrations are $\left[SO_{2(g)}\right]_e = 1M$, $\left[O_{2(g)}\right]_e = 0.5M$, $\left[SO_{3(g)}\right]_e = 4M$. Answer the following questions: (a) obtain the extent of the reaction (b) what were the concentrations of sulfur dioxide and sulfur trioxide that were reacted initially? (c) obtain an expression for the polynomial associated with the equilibrium (d) obtain the roots and demonstrate that you obtain the same extent value

4.49 Ammonium chloride is a versatile chemical salt used primarily as a nitrogen-rich agricultural fertilizer. When heated it decomposes into ammonia and hydrochloric acid according to

$$NH_4Cl_{(s)} \rightleftharpoons NH_{3(g)} + HCl_{(g)} \qquad K_p$$

After reaching equilibrium the total pressure in the container was found to be 5 atm. Obtain (a) The ICE chart using atmospheres as units of pressure (b) calculate the extent of reaction x at this temperature for the reaction in pressure units (c) calculate K_p at this temperature for the reaction (d) Obtain an expression for the polynomial associated with the equilibrium in terms of x and K_p

4.50 The reaction of elemental carbon with steam at high temperatures is called the water-gas generation or producing water gas (a mixture of carbon monoxide and hydrogen), whereas the reaction of carbon dioxide gas with water is called hydration forming carbonic acid

$$C_{(s)} + H_2O_{(g)} \rightleftharpoons CO_{(g)} + H_{2(g)} \qquad K\text{=6.89 at } 800°C$$

The initial mixture had a total pressure of 3atm and the total pressure when equilibrium is reached was 5atm. Obtain (a) The ICE chart using atmospheres as units of pressure (b) calculate the extent of reaction x at this temperature for the reaction in pressure units (c) Obtain an expression for the polynomial associated with the equilibrium in terms of x and K, and the initial pressure of each gas



4.51 Demonstrate step-by-step that the general formula for the polynomial associated with the equilibrium below in terms of K_c and the concentration of reactants and products, c_a, c_b, and c_c.

	$2A_{(g)}$	+	$B_{(g)}$	$\rightleftharpoons$	$2C_{(g)}$
Initial	c_a		c_b		c_c
Change	$-2x$		$-x$		$+2x$
Equilibrium	$c_a - 2x$		$c_b - x$		$c_c + 2x$

is given by

$$-4 \cdot Kx^3 + 4(c_a \cdot K + c_b \cdot K - 1)x^2 - (c_a^2 \cdot K + 4c_a \cdot c_b \cdot K - 4c_c)x + (c_a^2 \cdot c_b \cdot K - c_c^2) = 0$$

4.52 Demonstrate step-by-step that the general formula for the polynomial associated with the equilibrium below in terms of K_c and the concentration of reactants and products, c_a, c_b, and c_c.

	$2A_{(g)}$	$\rightleftharpoons$	$B_{(g)}$	+	$C_{(g)}$
Initial	c_a		c_b		c_c
Change	$-2x$		$+x$		$+x$
Equilibrium	$c_a - 2x$		$c_b + x$		$c_c + x$

is given by

$$(1 - 4K_c)x^2 + (c_b + c_c + 4K \cdot c_a)x + (c_b \cdot c_c - K \cdot c_a^2) = 0$$

4.53 Demonstrate step-by-step that the general formula for the polynomial associated with the equilibrium below in terms of K_c and the concentration of reactants and products, c_a, c_b, and c_c.

	$A_{(g)}$	$\rightleftharpoons$	$2B_{(g)}$
Initial	c_a		c_b
Change	$-2x$		$+2x$
Equilibrium	$c_a - 2x$		$c_b + 2x$

is given by

$$4x^2 + (4c_b + K)x + (c_b^2 - K \cdot c_a) = 0$$

4.54 Demonstrate step-by-step that the general formula for the polynomial associated with the equilibrium below in terms of K_c and the concentration of reactants and products, c_a, c_b, and c_c.

	$A_{(s)}$	$\rightleftharpoons$	$B_{(g)}$	+	$C_{(g)}$
Initial			c_b		c_c
Change			$+x$		$+x$
Equilibrium			$c_b + x$		$c_c + x$

is given by

$$x^2 + (c_b + c_c)x + (c_b \cdot c_c - K) = 0$$

4.55 Gaseous NOCl decomposes to form the gases NO and Cl_2 in an experiment in which 1.0 mol NOCl is placed in a 2.0-L flask. At 35°C, the equilibrium constant is 1.6×10^{-5}.

$$2\,NOCl_{(g)} \rightleftharpoons 2\,NOCl_{(g)} + Cl_{2(g)} \qquad K = 1.6 \times 10^{-5}$$

Answer the following questions (a) Obtain an expression for the exact polynomial $P(x)$ associated with the equilibrium, for those initial conditions, in terms of K_c, and the initial NOCL concentration c_a. Indicate the order and obtain the root x with chemical meaning. (b) Obtain an expression for the simplified polynomial $P_s(x)$ associated with the equilibrium, for those initial conditions, in terms of K_c, and the initial NOCL concentration c_a. (c) Compare x and x_s, and obtain the percent error of this approximation. Is the approximation valid?

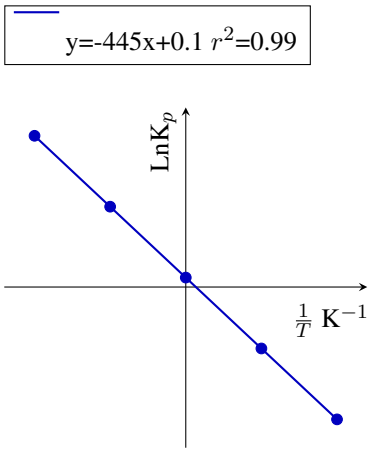
4.56 Water decomposes into hydrogen and oxygen through a process called electrolysis, and these gases can be burned together as a high-efficiency rocket propellant, following the reaction carried out at 2027°C

$$2\,H_2O_{(g)} \rightleftharpoons 2\,H_{2(g)} + O_{2(g)} \qquad K_p = 8.0 \times 10^{-3}$$

in an experiment in which 1.0atm mol H_2O is placed in a 1.0-L container. At 2027°C, the equilibrium constant K_p is 8.0×10^{-3}. Answer the following questions (a) Obtain an expression for the exact polynomial $P(x)$ associated with the equilibrium, for those initial conditions, in terms of K_c, and the initial water pressure p_a. Indicate the order and obtain the root x with chemical meaning. (b) Obtain an expression for the simplified polynomial $P_s(x)$ associated with the equilibrium, for those initial conditions, in terms of K_c, and the initial water pressure p_a. (c) Compare x and x_s, and obtain the percent error of this approximation. Is the approximation valid?

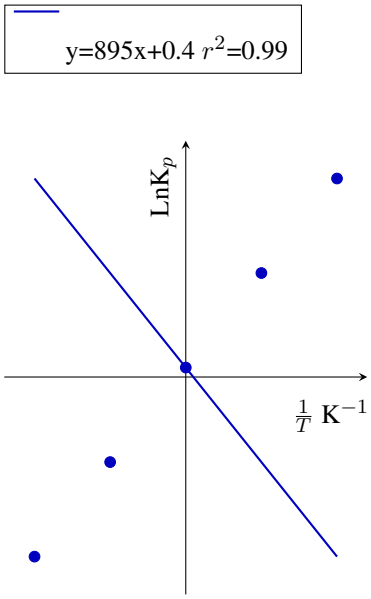
EFFECT OF TEMPERATURE ON K

4.57 In a set of experiments, we measure different K_p values at different temperatures. Answer the following questions:



(a) Is the reaction endothermic or exothermic? (b) Calculate the enthalpy of the reaction in KJ/mol

4.58 In a set of experiments, we measure different K_p values at different temperatures. Answer the following questions:



(a) Is the reaction endothermic or exothermic? (b) Calculate the enthalpy of the reaction in KJ/mol

4.59 Carry out a regression analysis using the following data to calculate the standard enthalpy of reaction (in kJ/mol units) for the reaction between gaseous hydrogen and iodine, indicating whether the reaction is endothermic or exothermic.

$H_{2(g)} + I_{2(g)} \rightleftharpoons 2\,HI_{(g)}$				
T (K)	298	500	700	800
K_p	794	160	54	48

4.60 Carry out a regression analysis using the following data to calculate the standard enthalpy of reaction (in kJ/mol units) for the decomposition of $BrCl_{(g)}$, indicating whether the reaction is endothermic or exothermic.

$2\,BrCl_{(g)} \rightleftharpoons Br_{2(g)} + Cl_{2(g)}$				
T (K)	300	500	800	1000
K_p	377	32	5	4



Answers **4.1** (a) False (b) True (c) True (d) True **4.2** (a) True (b) True (c) True (d) False **4.3** (a) $CH_{4(g)} + O_{2(g)} \longrightarrow CO_{2(g)} + H_2O_{(g)}$; $CH_{4(g)} + O_{2(g)} \longleftarrow CO_{2(g)} + H_2O_{(g)}$ (b) $2\,Mg_{(s)} + O_{2(g)} \longrightarrow 2\,MgO_{(s)}$;$2\,Mg_{(s)} + O_{2(g)} \longleftarrow 2\,MgO_{(s)}$ **4.4** (a) $BaCO_{3(s)} \longrightarrow Ba^{2+}_{(aq)} + {CO_3}^{2-}{}_{(aq)}$; $BaCO_{3(s)} \longleftarrow Ba^{2+}_{(aq)} + {CO_3}^{2-}{}_{(aq)}$ (b) $2\,H_{2(g)} + O_{2(g)} \longrightarrow 2\,H_2O_{(l)}$; $2\,H_{2(g)} + O_{2(g)} \longleftarrow 2\,H_2O_{(l)}$ **4.5** (a) More reactants (b) More products (c) More products **4.6** (a) same amount (b) More products **4.7** (a) right (b) left (c) center **4.8** (a) Center (b) left (c) right **4.9** (a) $K_c = \frac{[SO_3]^2}{[SO_2]^2\cdot[O_2]}$; $K_p = \frac{p^2_{SO_3}}{p^2_{SO_2}\cdot p_{O_2}}$ (b) $K_c = \frac{[CH_3OH]}{[H_2]^2\cdot[CO]}$;$K_p = \frac{p_{CH_3OH}}{p^2_{H_2}\cdot p_{CO}}$ (c) $K_c = \frac{[HCl]}{[C_2H_6]\cdot[Cl_2]}$; $K_p = \frac{p_{HCl}}{p_{C_2H_6}\cdot p_{Cl_2}}$ **4.10** (a) $[Ba^{2+}]\cdot[{CO_3}^{2-}]$ (b) $\frac{1}{[H_2]^2\cdot[O_2]}$ **4.11** (a) 1.6×10^{49} (b) 2.2×10^{24} (c) 6.4×10^{73} **4.12** (a) 142129 (b) 7.0×10^{-6} (c) 5.3×10^{7} **4.13** (a) 0.26 M (b) 0.46M **4.14** (a) $\frac{[SO_3]^2}{[SO_2]^2\cdot[O_2]}$ (b) 13.3 (c) mostly products **4.15** (a) K_c =17 ;K_p =10288 (b) K_c =2×10^{-3} ;K_p =0.05 **4.16** (a) K_c =0.243 ;K_p =5.977 (b) K_c =0.196 ;K_p =0.196 **4.17** (a) < – (b) < – **4.18** (a) $\longrightarrow$ (b) $\rightleftharpoons$ **4.19** $\longleftarrow$ **4.20** $\longrightarrow$ **4.21** (top) not in equilibriuml; evolved towards right; (bottom) not in equilibriuml; evolved towards left; **4.22** (top) not in equilibriuml; evolved towards left; (bottom) in equilibriuml **4.23** (a) $\longrightarrow$ (b) $\longleftarrow$ (c) $\longleftarrow$ **4.24** (a) $\longrightarrow$ (b) $\longleftarrow$ (c) $\longrightarrow$ **4.25** $\longleftarrow$ **4.26** $\longrightarrow$ **4.27** (a) $\longrightarrow$ (b) $\longleftarrow$ (c) $\longrightarrow$ **4.28** (a) $\longleftarrow$ (b) $\longrightarrow$ (c) $\longleftarrow$

4.29 (a)

	$2SO_{2(g)}$	+	$O_{2(g)}$	$\rightleftharpoons$	$2SO_{3(g)}$
I	2.5mM		0.5mM		0mM
C	$-2x$		$-x$		$2x$
E	$2.5-2x$		$0.5-x$		$2x$

(b)

	$2SO_{2(g)}$	+	$O_{2(g)}$	$\rightleftharpoons$	$2SO_{3(g)}$
I	0.5M		2.5M		2.5M
C	$-2x$		$-x$		$2x$
E	$0.5-2x$		$2.5-x$		2.5+$2x$

(c)

	$2SO_{2(g)}$	+	$O_{2(g)}$	$\rightleftharpoons$	$2SO_{3(g)}$
I	2M		0M		0.25M
C	$-2x$		$-x$		$2x$
E	$2-2x$		$-x$		0.25+$2x$

4.30 (a)

	$2BrCl_{(g)}$	$\rightleftharpoons$	$Br_{2(g)}$	$Cl_{2(g)}$
I	5atm		1atm	0atm
C	$-2x$		$+x$	$+x$
E	$5-2x$		$1+x$	x

(b)

	$2BrCl_{(g)}$	$\rightleftharpoons$	$Br_{2(g)}$	$Cl_{2(g)}$
I	1atm		1atm	5atm
C	$-2x$		$+x$	$+x$
E	$1-2x$		$1+x$	5+x

(c)

	$2BrCl_{(g)}$	$\rightleftharpoons$	$Br_{2(g)}$	$Cl_{2(g)}$
I	0atm		2atm	1atm
C	$-2x$		$+x$	$+x$
E	$0-2x$		$2+x$	1+x

4.31 Correct; **4.32** Incorrect;

	$2A_{(g)}$	+	$B_{(g)}$	$\rightleftharpoons$	$4C_{(g)}$	+	$2D_{(g)}$
I	1M		1M		1M		1M
C	$-2x$		$-x$		$+4x$		$+2x$
E	$1-1x$		$1-2x$		1+$4x$		1+$2x$

4.33 Incorrect;

	$3A_{(g)}$	+	$B_{(s)}$	$\rightleftharpoons$	$5C_{(g)}$	+	$2D_{(g)}$
I	1M				1M		1M
C	$-3x$				$+5x$		$+2x$
E	$1-1x$				1+$5x$		1+$2x$

4.34 correct; **4.35** (a) 0.05M (b) $6.4\times10^{-3}M$ **4.36** (a) $8.5\times10^{-3}M$ (b) 0.025M **4.37** (a) $4x^2+4\times10^{-2}x-5.5\times10^{-5}=0$ (b) $4x^2+4\times10^{-3}x-3.0\times10^{-5}=0$ (c) $4x^2+4\times10^{-3}x-9.2\times10^{-5}=0$ (d) $4x^2+3\times10^{-5}x-9.3\times10^{-5}=0$ (e) $4x^2+6.8\times10^{-2}x-0.204=0$ **4.38** (a) $4x^2+6.7\times10^{-3}x-2.7\times10^{-3}=0$ (b) $4x^2+2.7\times10^{-3}x-2.7\times10^{-3}=0$ (c) $4x^2+6.7\times10^{-3}x1.0\times10^{-6}=0$ (d) $4x^2+4.1\times10^{-3}x1.0\times10^{-6}=0$ (e) $4x^2+0.02x-9.5\times10^{-5}=0$ **4.39** (a) $-0.9x^2+2x-1=0$ (b) 0.76 (c) 1.46 is larger than the initial concentration, 1 (d) $[CO_{(g)}]_e = 0.24M$, $[H_2O_{(g)}]_e = 0.24M$, $[CO_{2(g)}]_e = 0.76M$, $[H_{2(g)}]_e = 0.76M$ **4.40** (a) $-0.9x^2+2.2x-1=0$ (b) 0.60 (c) 1.84 is larger than the initial concentration, 1 (d) $[CO_{(g)}]_e = 0.4M$, $[H_2O_{(g)}]_e = 0.4M$, $[CO_{2(g)}]_e = 2.6M$, $[H_{2(g)}]_e = 0.6M$ **4.41** (a) $-221.2x^2+112.6x-13.38=0$ (b) 0.189 (c) 0.319 as $[HI_{(g)}]_e$ would be negative (d) $[HI_{(g)}]_e = 0.122M$, $[H_{2(g)}]_e = 1.189M$, $[I_{2(g)}]_e = 0.689M$ **4.42** (a) $-221.2x^2+111.6x-13.88=0$ (b) 0.223 (c) 0.281 as $[HI_{(g)}]_e$ would be negative (d) $[HI_{(g)}]_e = 0.054M$, $[H_{2(g)}]_e = 0.723M$, $[I_{2(g)}]_e = 0.223M$

4.43 (a)

	$2HD_{(g)}$	$\rightleftharpoons$	$H_{2(g)}$	+	$D_{2(g)}$
I	0.6M		0M		0M
C	$-2x$		$+x$		$+x$
E	$0.6-2x$		x		x

(b) $-0.04x^2+0.624x-0.0936=0$ (c) 0.1514 (the root 15.44 does not make chemical sense) (d) $[HD_{(g)}]_e = 0.297M$, $[H_{2(g)}]_e = 0.151M$, $[D_{2(g)}]_e = 0.151M$

4.44 (a)

	$2HD_{(g)}$	$\rightleftharpoons$	$H_{2(g)}$	+	$D_{2(g)}$
I	0.33M		0.33M		0M
C	$-2x$		$+x$		$+x$
E	$0.33-2x$		$0.33+x$		x

(b) $-0.04x^2 + 0.68x - 0.0288 = 0$ (c) 0.0426 (the root 16.95 does not make chemical sense) (d) $\left[HD_{(g)}\right]_e = 0.248M$, $\left[H_{2(g)}\right]_e = 0.376M$, $\left[D_{2(g)}\right]_e = 0.0426M$

4.45 (a)

	$2SO_{2(g)}$	+	$O_{2(g)}$	$\rightleftharpoons$	$2SO_{3(g)}$
I	10M		12M		0M
C	$-2x$		$-x$		$+2x$
E	$10-2x$		$12-x$		$2x$

(b) 4M (c) $\left[SO_{2(g)}\right]_e = 2M$, $\left[O_{2(g)}\right]_e = 8M$, $\left[SO_{3(g)}\right]_e = 8M$ (d) 2 **4.46** (a)

	$2SO_{2(g)}$	+	$O_{2(g)}$	$\rightleftharpoons$	$2SO_{3(g)}$
I	2.5M		2.5M		2.5M
C	$-2x$		$-x$		$+2x$
E	$2.5-2x$		$2.5-x$		$2.5+2x$

(b) 1M (c) $\left[SO_{2(g)}\right]_e = 0.5M$, $\left[O_{2(g)}\right]_e = 1.5M$, $\left[SO_{3(g)}\right]_e = 4.5M$ (d) 54 **4.47** (a)

	$2SO_{2(g)}$	+	$O_{2(g)}$	$\rightleftharpoons$	$2SO_{3(g)}$
I	aM		bM		0M
C	$-2x$		$-x$		$+2x$
E	$a-2x$		$b-x$		$2x$

(b) $(-4)x^3 + (4a + 4b - \frac{4}{K_c})x^2 - (4ab + a^2 + \frac{4c}{K_c})x + a^2b - \frac{c^2}{K_c} = 0$ (c) 0.5M (d) $\left[SO_{2(g)}\right]_e = 1.1M$, $\left[O_{2(g)}\right]_e = 2M$ (e) $(-4)x^3 + (12.4)x^2 - (10.01)x + 2.42 = 0$ **4.48** (a) 0.5M (b) $\left[SO_{2(g)}\right]_e = 2M$, $\left[SO_{3(g)}\right]_e = 3M$ (c) $(-4)x^3 + (10)x^2 - (8)x + 2 = 0$ **4.49** (a)

	$NH_4Cl_{(s)}$	$\rightleftharpoons$	$NH_{3(g)}$	+	$HCl_{(g)}$
I			0atm		0atm
C			$-x$		x
E			$+x$		$+x$

(b) 2.5atm (c) 6.25 (d) $x^2 - K_p = 0$ **4.50** (a)

	$C_{(s)}$	+	$H_2O_{(g)}$	$\rightleftharpoons$	$CO_{(g)}$	+	$H_{2(g)}$
I			$a$atm		$b$atm		$c$atm
C			$-x$		x		x
E			$a-x$		$b+x$		$c+x$

(b) 2atm (c) $(1)x^2 + (\frac{1}{K} + b + c)x + (bc - \frac{a}{K}) = 0$ **4.51** **4.52** **4.53** **4.54** **4.55** (a) $P(x) = -4x^3 + 4Kx^2 - 4Kx + c_a^2 * k = 0$; order 3; 9.74×10^{-3} (b) $P_s(x) = 4x^3 - K \cdot c_a^2 = 0$; order 3; 1.00×10^{-2} (c) 3% **4.56** (a) $P(x) = -4x^3 + 4Kx^2 - 4Kx + c_a^2 * k = 0$; order 3; 1.07×10^{-1} (b) $P_s(x) = 4x^3 - K \cdot c_a^2 = 0$; order 3; 1.26×10^{-1} (c) 18% **4.57** (a) Endothermic (b) $+3.7\ KJ/mol$ **4.58** (a) Exothermic (b) $-7.4\ KJ/mol$ **4.59** 11KJ/mol; endothermic **4.60** 18KJ/mol; endothermic